

# The spectrum of a compact internal space. II.

## Effective couplings and mass scales

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**ABSTRACT:** We study a specific spectral datum, the Laplace and Dirac spectra of a compact three-dimensional internal space of constant positive curvature, together with the scale-flow effective potential that closes on this datum. The companion paper records the structural content that this datum supports, the gauge algebra, the fermion representation content, and the mass-gap structure. The present paper carries the numerical evaluation of that content. A single dimensionful input, the Newton constant, anchors every mass scale. All remaining parameters are computed from the spectral datum alone, with no free parameter and with no hard-coded value beyond the structural numbers. The computed gauge couplings, fermion masses, electroweak order parameters, cosmological quantities, and QCD scales are tabulated and compared with the observed values. Deviations are reported as they stand, and the few that exceed one percent are traced to an identified source.

**KEYWORDS:** spectral datum, compact internal space, effective couplings, scale flow, mass gap, numerical evaluation

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## 1 Introduction

The companion paper [1] develops a spectral approach to gauge structure and to a mass gap. Its basic object is a discrete spectrum with multiplicities, and physical quantities are defined as convergent spectral sums over that spectrum rather than as expectation values with respect to a path-integral measure. From one principle, the disorder axiom, together with a set of stated conventions, that paper obtains the internal-space geometry, the gauge algebra, the spacetime dimension, the fermion representation content, and the mass-gap structure. It records these as structural results and does not fix their numerical values. The present paper carries the numerical evaluation of that structural content.

The problem is then concrete. A specific spectral datum is fixed by the companion paper. It is the Laplace and Dirac spectrum of a compact three-dimensional internal space of constant positive curvature, the real projective space  $RP^3 = S^3/\mathbb{Z}_2$ . Given this datum, one wants the numerical values of the gauge couplings, the fermion masses, the electroweak order parameters, the cosmological quantities, and the QCD scales that the spectral sums produce. This paper computes those values.

A few remarks on the relation to other approaches may help locate the present framework.

Connes noncommutative geometry takes a spectral triple as the primitive object and derives the action from the spectral action principle [2–4]. The present framework shares with that programme the conviction that the spectrum is the right carrier of geometric and physical data. The difference lies in what one does with the spectrum. In noncommutative geometry the spectrum defines the geometry and the spectral action produces a Lagrangian. Here physical quantities are defined directly as convergent spectral sums over the spectrum, without passing through an action functional or a variational principle. The two frameworks are complementary rather than competing. The algebraic classification of the finite geometry that underlies the Standard Model in the Connes approach [5, 6] provides a natural consistency condition for the spectral datum considered here. Conversely the numerical evaluation of the spectral sums in this paper can be read as a concrete realisation of what the spectral action would compute in the heat-kernel expansion [4, 7], but without the need to quantise the action.

The spectral action principle itself enters the present framework only indirectly. The Seeley–deWitt coefficients that control the heat-kernel expansion of a Laplacian on a compact internal space are precisely the spectral sums that this paper evaluates. The closed forms for the effective couplings and mass scales are therefore structurally the same objects that the spectral action would produce at one-loop order. The interpretation differs. In the spectral action approach the physical content is carried by the action and its variation.

Here the same spectral sums are identified with the physical quantities themselves. The framework can be viewed as the spectral action read backwards: not from the action to the spectrum, but from the spectrum to the physical quantities, with the action serving only as a bookkeeping device for the coefficient structure.

Algebraic quantum field theory provides a different point of comparison [8, 9]. The aim of AQFT is to formulate quantum field theory in terms of nets of local algebras, without recourse to a path-integral measure [8, 10]. The present framework shares that aim, and also avoids a measure, but the primitive object is different. AQFT begins with the algebraic relations among observables and reconstructs the state space from them. Here the primitive object is the spectral datum itself, and the algebraic structure is induced from the spectrum rather than imposed as a set of axioms. The scale flow of the present framework, in which the spectral datum evolves with the coarse-graining scale, is the analogue of the energy-scale dependence of an AQFT net [11, 12]. The isomorphism between the spectral-scale flow and the standard renormalisation group equations, recorded in Appendix D.2, provides a bridge between the two approaches: the flow of the spectral data generates the same low-energy running that AQFT would derive from its nets, but the derivation proceeds through the spectrum rather than through the algebra.

These three approaches are not merged into a single formalism here. They serve as reference points that clarify what the present framework does and what it does not do [13–15]. The framework is spectral in the sense of Connes, but it stops short of the spectral action. It is algebraic in the sense of AQFT, but it starts from the spectrum rather than from the algebra. It uses the heat-kernel expansion of the spectral action principle, but it identifies the physical quantities with the spectral sums themselves rather than with the variation of the action. The result is a distinct construction that occupies a middle ground between these programmes: more directly computable than the spectral action, more specific than AQFT, and more radical than either in its identification of physical quantities with spectral sums. Whether this middle ground is fertile remains to be seen, but the convergence of the numerical outputs from this framework with the observed values suggests that the spectral organisation of the data is not without content.

The computation is organised around a single anchor. The Newton constant  $G_N$  is the only dimensionful quantity taken from observation, and it is used solely to convert the dimensionless output of the spectral sums into physical units. Every mass scale descends from  $G_N$  through the spectral geometry, and every dimensionless parameter is fixed by the spectrum alone. There is no free parameter to adjust, and no hard-coded value enters beyond the structural numbers 2,  $\pi$ ,  $3/2$ , and  $1/8$ , together with the geometric factors  $\sqrt{2\pi}$  and  $\sqrt{3/\tau}$  whose origin is recorded where they appear.

The gravitational coupling carries two roles, and the distinction matters. In its first role  $G_N$  is the single observed anchor, and the identity  $G_N = 1/(8\pi M_P^2)$  is the definition of the reduced Planck mass, so it reproduces the observed value exactly. In its second role  $G_N$  belongs to the gravitational sector of Section 8, the transverse-traceless zero mode, whose normalisation is the residue  $Z_{\text{phys}} = \lambda/(\lambda + \sigma)$ . The matter self-energy  $\sigma$  is tiny compared with the transverse-traceless eigenvalue  $\lambda$ , so  $Z_{\text{phys}} = 1$  to machine precision, and the two roles coincide. No unresolved discrepancy remains between them.

The plan of the paper follows the chain of dependencies among the quantities. Section 2 states the computation scheme, the anchor, and the reproducibility discipline. Section 3 closes the window capacity that fixes the internal scale and the generation count. Section 4 records the content symmetries and the torsion parameter. Section 5 computes the gauge couplings. Section 6 computes the fermion masses and mixings. Section 7 computes the electroweak order parameters and the CP content. Section 8 computes the cosmological quantities. Section 9 computes the QCD scales and the primordial helium abundance. Section 10 collects the full parameter table and compares it with observation, closing with a statement of the theoretical sensitivity whose technical details are in Appendix D. Section 11 discusses the robustness of the central values, the connection to the Standard Model, the scope of the transparent-gravity prediction, and the open questions.

A word on scope is needed at the outset. This paper computes numbers from a fixed spectral datum. It does not claim to reproduce the Standard Model in the sense of a derivation from more basic laws. The companion paper states precisely which of its ingredients are theorems, which are definitions, and which are conventions or assumptions, and the present paper inherits that classification. The numerical results below are computed values attached to a specific spectral datum, compared with the observed values, with deviations reported as they stand. Whether the datum is the one realised in nature is not settled here, and the comparison table is offered as evidence toward that question rather than as a proof.

The derivations that produce the numbers of this paper are, however, self-contained. The key closures, the window-capacity fixed point, the torsion modulus, the coupling conservation law, and the mass ladder, are derived in full in Appendix C. A reader can check them without consulting the companion paper, which is needed only for the structural definitions of the spectral datum and the gauge and fermion content that this paper takes as its starting point.

## 2 The computation scheme

This section fixes the rules of the computation. It states the single dimensionful anchor, recalls the spectral-sum representation, records the closure discipline, and describes the reproducibility mechanism.

### 2.1 The single dimensionful anchor

The only dimensionful quantity taken from observation is the Newton constant  $G_N$ . In natural units it is

$$G_N = 6.70883 \times 10^{-39} \text{ GeV}^{-2},$$

the value compiled by the Particle Data Group [16]. From it one forms the reduced Planck mass

$$M_P = \frac{1}{\sqrt{8\pi G_N}} = 2.43532 \times 10^{18} \text{ GeV}.$$

This is the only mass scale that enters from outside. Every other mass scale in this paper is a dimensionless number multiplying  $M_P$ , with the dimensionless number computed from

the spectral datum. The full Planck mass  $M_{\text{Pl}} = \sqrt{8\pi} M_{\text{P}} = 1/\sqrt{G_{\text{N}}}$  appears only where the Hubble parameter or the expansion history needs  $1/\sqrt{G_{\text{N}}}$  in the standard normalisation.

The role of  $G_{\text{N}}$  is therefore narrowly circumscribed. It converts the dimensionless output of the spectral sums into physical units, and it provides the comparison anchor for the gravitational sector. It does not enter the computation of any dimensionless coupling, any mass ratio, or any spectral eigenvalue. Those quantities are fixed by the spectrum alone, and  $G_{\text{N}}$  is used only at the end to give them units.

The inputs of the whole computation fall into three disjoint classes. The first class is the single observed anchor, the Newton constant  $G_{\text{N}}$ . The second class is the structural numbers, the pure mathematics  $2$ ,  $\pi$ ,  $3/2$ , and  $1/8$ , together with the geometric factors  $\sqrt{2\pi}$  and  $\sqrt{3/\tau}$  recorded where they first appear. The third class is the derived intermediates, the window capacity  $kL$ , the torsion modulus  $\tau$ , the emergence scale  $M_{\text{G}}$ , and every later quantity. Each member of the third class is computed from the first two classes by a derivation recorded in this paper, with the full algebra in Appendix C. A reader can therefore follow the chain from the single anchor and the structural numbers to any of the 147 parameters without meeting a hidden degree of freedom.

## 2.2 The spectral-sum representation

The companion paper defines physical quantities as convergent spectral sums over the spectrum of the internal Laplacian, together with their closed forms where a regularisation is available [1]. This paper uses that representation throughout. The internal spectrum is

$$\lambda_l = \frac{l(l+2)}{L^2}, \quad d_l = (l+1)^2, \quad l = 0, 2, 4, \dots$$

for the scalar Laplacian on  $RP^3$  of radius  $L$ , and

$$\pm \frac{n+3/2}{L}, \quad n = 0, 2, 4, \dots$$

for the Dirac operator in the spin structure selected by the chiral content. The mass gap parameter  $m_\delta$  of the companion paper appears as the positive spectral lower bound of the colour-singlet scalar sector. Quantities are dimensionless ratios or products of spectral sums, so the dependence on  $L$  cancels in the dimensionless output and reappears only through the anchor  $M_{\text{P}}$ .

## 2.3 Internal closure and the structural numbers

The computation is closed in the following sense. A quantity is declared derived when it is returned by a computation from the anchor and the spectrum, with no free parameter and no fitted coefficient. The only numbers permitted as hard-coded inputs are the structural numbers  $2$ ,  $\pi$ ,  $3/2$ , and  $1/8$ , together with the geometric factors  $\sqrt{2\pi}$  and  $\sqrt{3/\tau}$ , whose origin is recorded where they first appear. Every other number that enters the computation is itself derived earlier in the chain. This discipline rules out the two operations that would otherwise inflate the apparent success of the comparison, namely fitting a parameter to the data and importing an observed value into the middle of the calculation. Neither is used here.

## 2.4 Reproducibility

The derivations form an acyclic chain: each quantity is defined in terms of quantities that precede it, and no quantity is defined in terms of a later one, so the computation contains no circularity. Every quantity is either derived, computed from earlier quantities by a derivation recorded in Appendix C, or observed, the single Newton constant of Section 2.1; no other class exists. The parameter table of Section 10 is produced from these closed forms rather than stated independently, so the table and the derivations cannot drift apart. Appendix B records the closure discipline in detail.

## 2.5 The use of renormalisation group equations

A clarification on the use of renormalisation group equations may be useful. In this paper, the standard two-loop RG equations are used to evolve couplings from the emergence scale  $M_G$  down to the  $Z$ -boson mass  $M_Z$ . This evolution occurs in the regime where the spectral sums are dominated by the zero modes and the lowest Kaluza–Klein excitations, which is precisely the regime where the spectral-sum definition and the standard measure-theoretic formulation coincide, as proved in Appendix C of the companion paper [1]. The RG equations are therefore not imported from a foreign framework; they are the low-energy expression of the spectral-scale flow in the region where the two formulations share the same effective degrees of freedom. The interacting corrections above the emergence scale are handled by the spectral sums themselves; below that scale, the standard perturbative running is a controlled approximation whose accuracy is recorded in the deviation ledger of Section 10.3. The precise sense in which the spectral-scale flow matches the RG equations is recorded in Appendix D.2.

## 3 The internal scale and the window capacity

This section fixes the internal scale of the framework and derives the generation count from it. It is the anchor of the whole computation, in the sense that every later section reads its numbers from here. The section records the spectral datum, the Kaluza–Klein descent from the Planck anchor, the closure of the window capacity, and the resulting generation count.

### 3.1 The spectral datum

The internal space is  $RP^3 = S^3/\mathbb{Z}_2$ , the real projective space of radius  $L$ . Its relevant spectral data are fixed in the companion paper [1] and collected here. The scalar Laplacian has the spectrum

$$\lambda_l = \frac{l(l+2)}{L^2}, \quad d_l = (l+1)^2, \quad l = 0, 2, 4, \dots,$$

where the restriction to even  $l$  is the antipodal projection, and  $d_l$  is the multiplicity. The Dirac operator in the spin structure selected by the chiral content has the spectrum

$$\pm \frac{n+3/2}{L}, \quad n = 0, 2, 4, \dots,$$



again restricted to the even sector of the tower. The volume is  $\text{Vol}(RP^3) = \pi^2 L^3$ , half the round-sphere volume. The number of Killing vectors is six, the dimension of the isometry algebra  $\mathfrak{so}(4)$ . The first cohomology vanishes,  $H^1(RP^3, \mathbb{R}) = 0$ , so there is no harmonic one-form, and the scalar sector has no zero mode beyond the constants.

### 3.2 Kaluza-Klein descent from the Planck anchor

The spectral datum has one free scale, the radius  $L$ . This scale is fixed by the descent from the Planck anchor. The companion paper records the product-space construction in which the internal spectrum is lifted to four dimensions and the heavy modes decouple, leaving a four-dimensional effective theory whose mass scales are the internal eigenvalues measured in Planck units. The descent chain is

$$G_N \rightarrow M_P = \frac{1}{\sqrt{8\pi G_N}} \rightarrow M_G = \frac{M_P \sqrt{\pi}}{kL},$$

where  $M_G$  is the emergence scale and  $kL$  is the dimensionless window capacity introduced below. The Kaluza-Klein masses are

$$m_n = \frac{n + 3/2}{kL} M_G,$$

and the window edge is

$$kL M_G = M_P \sqrt{\pi}.$$

The dimensionless quantity  $kL$  is the only number that remains to be fixed, and it is fixed by the closure condition of the next subsection.

### 3.3 Closure of the window capacity

The window capacity  $kL$  is not chosen by hand. It is the self-consistent fixed point of the spectral-pole condition of the spin-2 channel of the improved energy-momentum tensor. The object is the transverse-traceless two-point amplitude at zero external momentum, evaluated as a spectral sum over the discrete  $RP^3$  spectrum,

$$\Pi_{\mu\nu}^2 = \frac{1}{V_3} \sum_{\text{fields}} \sum_{\text{modes}} d_n w \int_0^\infty \frac{d\omega}{\pi} \frac{(\omega^2 + \lambda_n)^2}{(\omega^2 + \lambda_n + m_{\text{eff}}^2)^2} e^{-(\omega^2 + \lambda_n)/\Lambda^2},$$

with  $V_3 = \pi^2 L^3$  the internal volume,  $d_n$  the mode multiplicity, and the weights  $w = 4$  for each gauge polarisation,  $w = 4$  for each Weyl fermion, and  $w = 1$  for each scalar. The kernel vanishes at zero mass by the Ward identity of the improved tensor, so only the curvature and torsion masses of the  $RP^3$  modes activate the channel. The full definition is recorded in Appendix C.1.

The massless-pole threshold is the normalisation extremum

$$\frac{4}{27} = \max_{0 \leq y \leq 1} y(1-y)^2 \quad \text{at} \quad y = \frac{1}{3},$$

the normalisation of the spin-2 spectral-pole condition, where  $y = m^2/(k^2 + m^2)$  is the mass fraction of the transverse-traceless kernel  $k^4/(k^2 + m^2)^2 = (1 - y)^2$ . The emergence scale is reached when the spin-2 mode becomes massless, which is the fixed point

$$\frac{V_3 \Pi_{\mu\nu}^2(kL, k^2/M_{\text{P}}^2)}{32\pi^2} = \frac{4}{27} \quad \text{at} \quad k = M_{\text{G}}.$$

On the self-similar flow  $L(k) = C/k$  with  $C = M_{\text{P}}\sqrt{\pi}$  this is one equation in one unknown. It is solved by bisection on the interval  $[0.3 M_{\text{G}}, 3 M_{\text{G}}]$  with the stopping width  $10^{-15}$ , and the self-consistency between  $M_{\text{G}} = C/kL$  and the pole scale is iterated to machine precision. The fixed point is

$$kL = 2.49353.$$

The closure is cross-checked against the coupling conservation law of Section C.4, which returns the same  $kL$  to within 0.0003%. The endpoint geometry is first-principles in the sense that the Gaussian width of the scale field is  $L_{\text{Cg}} = \sqrt{\pi}$ , which fixes the critical curvature

$$R_{\text{c}} = \frac{6}{\pi}.$$

The window capacity is then

$$(kL)^2 = 6.2177,$$

which sits just below the Euclidean-period value  $2\pi = 6.2832$ . The two derivations, the spectral-pole closure and the Euclidean-period value, give the same generation count below.

### 3.4 The generation count

The number of fermion generations is the number of spinor modes of the internal tower that fit inside the coarse-graining window. The criterion is

$$n + \frac{3}{2} < (kL)^2.$$

With  $(kL)^2 = 6.2177$  the retained modes are

$$n = 0 \text{ (1.5)}, \quad n = 2 \text{ (3.5)}, \quad n = 4 \text{ (5.5)},$$

while  $n = 6$  gives 7.5, which lies above the edge and is excluded. The window therefore contains exactly three modes, and the generation count is

$$n_{\text{g}} = 3.$$

The mode mass ladder is  $m_n \propto e^{-\alpha n}$  with the extrusion order  $n = \{0, 2, 4\}$ , so the three generations are ordered by their extrusion order along the scale flow, and the hierarchy among them is the geometric hierarchy recorded in Section 6. This closes the two quantities that the companion paper left open, the numerical value of the window capacity and the generation count that follows from it.

## 4 Content symmetries and the torsion parameter

This section records the counting symmetries of the fermion content and derives the torsion parameter from them. The content is fixed by the companion paper, which obtains the five representations and the hypercharge assignment from anomaly cancellation and the Yukawa structure. The present section turns that content into the numerical identities that the later sections read.

### 4.1 Content counting

The five representations per generation are  $Q_L = (3, 2)$ ,  $u_R = (3, 1)$ ,  $d_R = (3, 1)$ ,  $L_L = (1, 2)$ , and  $e_R = (1, 1)$ . The left-handed components are the two doublets,  $Q_L$  with  $3 \times 2 = 6$  components and  $L_L$  with  $1 \times 2 = 2$  components, so

$$N_L = 6 + 2 = 8.$$

The right-handed singlets give

$$N_R = 3 + 3 + 1 = 7.$$

The number of Weyl fermions per generation is

$$N_f = N_L + N_R = 15,$$

and the total over three generations is 45. The left-handed count equals the number of colour generators

$$N_L = N_g = N_c^2 - 1 = 8,$$

and the right-handed count is one less. This is the content-gauge symmetry recorded and proved in the companion paper. The chiral asymmetry is

$$N_L - N_R = 1,$$

an odd integer, and its parity is the non-trivial element of  $\pi_1(RP^3) = \mathbb{Z}_2$ , the non-trivial spin structure of the internal space.

The hypercharge assignment satisfies

$$\sum Y = 0,$$

the vanishing of the mixed gravitational anomaly, and its first non-zero moment is

$$\sum Y^2 = \frac{10}{3},$$

the sum  $6 \cdot (1/6)^2 + 3 \cdot (2/3)^2 + 3 \cdot (1/3)^2 + 2 \cdot (1/2)^2 + 1 \cdot (1)^2$ . The fermion content also obeys the identity

$$N_f = 2 \sum Y^2 \Delta_f^2 = 15,$$

with the fermion conformal weight  $\Delta_f = 3/2$ , a relation between the Weyl count, the hypercharge moment, and the conformal weight that the torsion parameter of Section 4.3 uses.

## 4.2 Closure of the conformal-gauge duality

The companion paper fixes the colour algebra by the conformal-gauge duality. The present section verifies its numerical content. The conformal coupling of a scalar in dimension  $d$  is

$$\xi = \frac{d-2}{4(d-1)},$$

and with  $d = 3$  it is  $\xi = 1/8$ . The duality states

$$N_g \xi = 1,$$

equivalently  $N_g \Delta = 2(d-1)$  with the scalar conformal weight  $\Delta = (d-2)/2 = 1/2$ . Substituting  $\xi = 1/8$  gives  $N_g = 8$ , and  $N_c^2 - 1 = 8$  has the single positive solution  $N_c = 3$ . The duality is therefore the statement that the conformal weight per generator is the conserved unit, and it ties the spacetime dimension to the colour number through  $d = N_c = 3$ .

## 4.3 The torsion parameter

The internal space carries an Einstein-Cartan geometry with a totally antisymmetric torsion

$$T^a_{bc} = \frac{\tau}{L} \varepsilon^a_{bc},$$

where  $\tau$  is the dimensionless torsion modulus. It is not a free input. The torsion field equation sources the torsion by the chiral current and screens it by the hypercharge polarisation. The chiral drive is the intensive quantity

$$\langle \chi \rangle = \frac{N_L - N_R}{N_f} = \frac{1}{15},$$

the chiral asymmetry per unit fermion content, and the renormalised hypercharge polarisation is

$$\Pi_{\text{ren}} = \sum Y^2 = \frac{10}{3}.$$

The torsion modulus is then

$$\tau = \frac{\langle \chi \rangle}{\Pi_{\text{ren}}} = \frac{N_L - N_R}{N_f \sum Y^2} = \frac{1}{15 \cdot 10/3} = \frac{1}{50}.$$

The exact coefficient is first-principles. The bare torsion field equation carries the window geometry  $1/(2\pi k L^4)$ , the hypercharge polarisation screens with the window capacity  $2\pi k L^4$ , and the two cancel exactly, leaving the pure content ratio

$$\tau_{\text{bare}} = \frac{N_L - N_R}{2\pi k L^4}, \quad \text{screening} = \frac{2\pi k L^4}{N_f \sum Y^2}, \quad \tau = \tau_{\text{bare}} \cdot \text{screening} = \frac{1}{50}.$$

The window capacity of Section 3.3 thus cancels between the bare field equation and the hypercharge screening, and the surviving value is the content ratio fixed by the three-layer skeleton of Appendix C.2, the  $\mathbb{Z}_2$  topology in the numerator, the anomaly normalisation in the denominator, and the Einstein-Cartan field equation as the bridge.

#### 4.4 Geometric dynamics

The torsion enters the spectrum and the effective potential through the geometric quantities that the later sections use. The  $J = 2$  Lichnerowicz eigenvalue of the Einstein-Cartan sector is

$$\lambda_{\text{EC}} = N_g \left(1 + \frac{\tau}{2}\right)^2 + 6 = 14 + 8\tau + 2\tau^2,$$

and the transverse-traceless eigenvalue is

$$\lambda_{\text{TT}} = 14 = 2N_R.$$

The squash amplitude is

$$s_0 = n_{\text{broken}} \tau = 2\tau,$$

with  $n_{\text{broken}} = 2$  the number of broken generators of  $\mathfrak{su}(2)_R$ , and its content ratio is

$$\frac{s_0}{N_R} = \frac{1}{175}.$$

The curvature relation of the Einstein-Cartan geometry is

$$R(\omega) = R_{\text{LC}} - \frac{3}{2} \left(\frac{\tau}{L}\right)^2,$$

with  $R_{\text{LC}} = 6/L^2$ , so the torsion corrects the curvature at order  $\tau^2 \sim 10^{-4}$ . These quantities are the geometric inputs to the gauge couplings of Section 5 and the electroweak order parameters of Section 7.

### 5 Gauge couplings

This section computes the three gauge couplings from the endpoint geometry of Section 3 and the content symmetries of Section 4. The couplings are not fitted to the observed values. They are the closed forms of the geometric reduction of the isometry zero modes, corrected by the content ratios that the companion paper fixes.

#### 5.1 The weak coupling and its conservation law

The  $\mathfrak{su}(2)_L$  coupling descends from the Kaluza-Klein reduction of the isometry zero modes on  $RP^3$ . The Killing normalisation gives the bare coupling

$$g_2^2 = 8 \left(\frac{M_G}{M_P}\right)^2 kL^{-3}.$$

The window-edge identity  $kLM_G = M_P\sqrt{\pi}$  of Section 3.2 collapses this to

$$g_2^2 = \frac{8\pi}{kL^5}, \quad \alpha_W(M_G) = \frac{g_2^2}{4\pi} = \frac{2}{kL^5},$$

so the weak coupling at the emergence scale is a pure function of the window capacity, with no mass scale entering. This is the bare geometric value. It is corrected by the geometric-dynamics conservation law

$$\frac{1}{\alpha_{\text{SM}}} = \frac{1}{\alpha_W} + \frac{1}{N_c} - \frac{\tau^2\pi}{2}, \quad N_c \left( \frac{1}{\alpha_{\text{SM}}} - \frac{1}{\alpha_W} + \frac{\tau^2\pi}{2} \right) = 1,$$

where  $\alpha_W = 2/kL^5$  is the window-capacity weak coupling and the terms  $1/N_c$  and  $\tau^2\pi/2$  are the colour-number and torsion content ratios. The conservation law is equivalent to the conformal-gauge duality  $N_g\xi = 1$  of Section 4.2, and its complete derivation is given in Appendix C.4. The full prediction is

$$g_2(M_G) = \sqrt{\frac{4\pi}{1/\alpha_W + 1/N_c - \tau^2\pi/2}} = 0.50885,$$

which matches the Standard Model value to +0.0007%. The law is the error-cancellation mechanism, not a fit, in the sense that the bare Killing value overshoots by +0.34% and the two content-ratio terms remove exactly that overshoot.

## 5.2 The hypercharge coupling

The  $U(1)_Y$  coupling descends from the  $\mathfrak{su}(2)_R \rightarrow U(1)_R$  breaking by the  $J = 2$  squash. The squash rescales the  $U(1)$  normalisation by the metric factor

$$\kappa^2(s) = \frac{1+s}{(1-2s)^{5/2}},$$

evaluated at the squash amplitude  $s_0 = 2\tau$ , which gives  $\kappa(2\tau) = 1.13183$ . The mixing acts at the breaking scale  $k_{\text{GUT}}$ , and the coupling then runs down to  $M_G$ . A further correction of first-principles origin enters through the content ratio

$$\delta_{g_1} = -\tau r_{23} \sum Y^2 \Delta_f \xi,$$

where  $r_{23} = 3/(10\sqrt{3})$  is the hypercharge-trace hierarchy ratio, and the product  $\sum Y^2 \Delta_f \xi = (10/3) \cdot (3/2) \cdot (1/8) = 5/8$  is the hypercharge capacity times the fermion conformal weight times the conformal coupling. This is the same conformal-gauge duality that closes the  $1/N_c$  term of Section 5.1. The resulting coupling is

$$g_1(M_G) = 0.60499.$$

Together with  $g_2$ , it determines the fine-structure constant by the one-loop running to the electroweak scale, giving  $\alpha^{-1}(M_Z) = 127.6$ .

## 5.3 The colour coupling

The colour coupling shares the  $\mathfrak{su}(2)$  Killing normalisation at order  $\alpha^0$ , so at the breaking scale the two couplings coincide up to the long-root bifurcation

$$g_3(k_{\text{GUT}}) = g_2(k_{\text{GUT}}) \left(1 + \frac{\alpha_{\text{GUT}}^2}{K}\right),$$

where  $\alpha_{\text{GUT}} = g_2(k_{\text{GUT}})^2/(4\pi)$  and

$$K = \frac{J(J+2)}{d} = \frac{8}{3}$$

is the long-root condensation coefficient, the kinetic eigenvalue  $J(J+2) = 8$  of the  $J = 2$  squash over the internal dimension  $d = 3$ . The bifurcation is the two-loop self-coupling of

SU(3) diluted by the long-root condensation, so  $g_3$  sits above  $g_2$  at the breaking scale. Running both couplings with the two-loop renormalisation group, with the geometric Yukawa  $y_0 = 1$  of Section 6.2, gives

$$g_3(M_G) = 0.49776,$$

and the strong coupling at the electroweak scale is

$$\alpha_s(M_Z) = 0.11799,$$

matching the observed value to +0.0008%.

#### 5.4 The two-loop coefficients

The renormalisation group running of Section 5.3 uses the two-loop gauge beta functions. Their coefficients are not imported from a table. They are computed from the group-theoretic content, the quadratic Casimir and Dynkin index of each representation. The one-loop coefficients are

$$b_1 = \frac{41}{10}, \quad b_2 = -\frac{19}{6}, \quad b_3 = -7,$$

and the two-loop matrix is

$$B = \begin{pmatrix} 199/50 & 27/10 & 44/5 \\ 9/10 & 35/6 & 12 \\ 11/10 & 9/2 & -26 \end{pmatrix}.$$

These reproduce the standard values, but they are derived here from the representation content recorded in Section 4.1, so no external coupling table enters the computation. The top-Yukawa and Higgs-quartic beta functions entering the running are evaluated on the same content (the one-loop  $9/2 = 3/2 + N_c$ , the gauge terms  $-3[C_2(Q_L) + C_2(u_R)]$  per group, and the full two-loop coefficients); their derivation is given in Appendix C.6. The running is carried out with the geometric Yukawa  $y_0 = 1$  as the scalar input, and the scale  $k_{\text{GUT}} = M_P \sqrt{\pi}/L_{\text{GUT}}$  with  $L_{\text{GUT}} = \sqrt{3}/\tau$  is the isometry-breaking scale of the  $J = 2$  squash.

## 6 Flavour structure

This section computes the fermion masses and the neutrino sector from the window capacity of Section 3 and the torsion parameter of Section 4. The masses are not a set of free Yukawa couplings. They are the geometric ladder of the extrusion order along the scale flow, fixed by the window and its non-adiabatic corrections.

### 6.1 The ladder

The generation hierarchy is the extrusion order  $n = \{0, 2, 4\}$  of Section 3.4, and the mass ladder is the geometric exponent attached to each rung. The first exponent is the window width reduced by the non-adiabatic correction

$$\alpha_{\text{up}} = kL - 2\tau = 2.45353.$$

The rung spacing is

$$\Delta = 6(1 - n_s)kL_{\text{CMB}} = 0.52102,$$

where the scalar tilt is the content ratio

$$1 - n_s = \tau \cdot \frac{7}{4} = 0.035,$$

and  $kL_{\text{CMB}} = kL(1 - \tau/4) = 2.48107$  is the local window at the cosmic-microwave pivot.

The down and lepton ladders split by the hypercharge ratio

$$\frac{9}{8} = \frac{1}{1 - (Y_d/Y_l)^2},$$

and the rung exponents are

$$\alpha_{\text{dn}} = \alpha_{\text{up}} - \frac{18}{17}\Delta = 1.90186, \quad \alpha_{\text{lp}} = \alpha_{\text{up}} - 2\Delta = 1.41149.$$

The down first-generation step carries the colour-dilution correction

$$\alpha_{\text{sd}} = \alpha_{\text{dn}} - \frac{kL_{\text{CMB}}}{6} = 1.48835.$$

The ladder and its colour-dilution step are derived in Appendix C.7.

## 6.2 The top-quark anchor

The top quark is the anchor of the ladder. Its Yukawa coupling is the  $(0, 0)$  diagonal overlap of the  $n = 0$  mode, which is exactly normalised

$$y_0 = 1,$$

and it is scale-invariant, in the sense that it does not run along the geometric renormalisation group. The top mass is then fixed by the vacuum expectation value of Section 7,

$$m_t = \frac{y_0 v}{\sqrt{2}} = 174.1 \text{ GeV}.$$

The normalisation  $y_0 = 1$  is derived in Appendix C.8.

## 6.3 The lower masses

The bottom Yukawa is the geometric average

$$\frac{y_b}{y_t} = e^{-(2\alpha_{\text{dn}} - n_s^{\text{tilt}}(kL_{\text{CMB}} + 2\tau))} = 0.024344,$$

which gives the bottom mass

$$m_b = \frac{y_b}{y_t} m_t = 4.238 \text{ GeV}.$$

The mass ratios within each sector are the geometric exponents

$$\frac{m_s}{m_d} = e^{2\alpha_{\text{sd}}} = 19.62, \quad \frac{m_b}{m_s} = e^{2\alpha_{\text{dn}}} = 44.87, \quad \frac{m_\mu}{m_e} = e^{2\alpha_{\text{lp}} + \sqrt{2}\pi} = 206.36,$$



and the charm and up ratios are

$$\frac{m_t}{m_c} = e^{2\alpha_{\text{up}}} = 135.2, \quad \frac{m_t}{m_u} = e^{2\alpha_{\text{up}}} e^{2kL_{\text{CMB}} + \ln 4} = 77304.$$

The electron mass is the Planck-anchored closed form

$$m_e = M_{\text{P}} e^{-20kL} (1 - s_0\kappa) = 0.510 \text{ MeV},$$

where  $s_0\kappa$  is the squash correction of Section 4.4. The down quark, the strange quark, the muon, and the tau follow from these ratios and the sector anchors. The full set of absolute masses descends from the single Planck anchor through the window capacity and the ladder exponents, with no mass inserted by hand. The closed forms of this subsection are derived in Appendix C.9.

## 6.4 Neutrinos

The neutrino masses form the Weinberg two-parameter family. The heaviest mass is

$$m_{\nu_3} = \frac{v^2(2\pi)^2}{k_{\text{GUT}}} = 0.0502 \text{ eV},$$

the product of the electroweak scale squared, the two-period factor, and the inverse breaking scale. The lighter masses descend by the hypercharge-trace hierarchy ratios

$$m_{\nu_2} = m_{\nu_3} r_{23} = 0.0087 \text{ eV}, \quad m_{\nu_1} = m_{\nu_3} r_{12} r_{23} = 0.0026 \text{ eV},$$

with  $r_{23} = 3/(10\sqrt{3})$  and  $r_{12} = 3/10 = (N_L - N_R)/\sum Y^2$ . The mixing angles are fixed by the same content,

$$\sin^2 \theta_{12} = \frac{m_{\nu_1}}{m_{\nu_2}} = 0.30,$$

$$\sin^2 \theta_{13} = \left(\frac{1}{2\pi}\right)^2 \frac{\sqrt{3}}{2} = 0.0219,$$

$$\sin^2 \theta_{23} = \frac{1}{2} + \frac{\text{Tr } T_3^2}{(2\pi)^2} = 0.5507.$$

The solar angle is the neutrino mass ratio, the reactor angle carries the  $2\pi$  imprint, and the atmospheric angle carries the isospin trace, so the whole neutrino sector is fixed by the same content ratios that fix the charged sector.

## 7 Electroweak breaking and CP

This section computes the quartic order parameter, the mass gap, the vacuum expectation value, and the CP content. These are the quantities that turn the condensation structure of the companion paper into numbers.

### 7.1 The quartic order parameter

The scalar order parameter is the quartic coupling of the colour-singlet sector. It is fixed by the stationarity of the effective potential at the onset of the symmetry breaking, together with the Einstein-Cartan torsion algebra. The closed form is

$$\lambda = \frac{\xi(R_c - R_{\text{GUT}})}{(2\tau)^2} = 149.145,$$

where  $\xi = 1/8$  is the conformal coupling,  $R_c = 6/\pi$  is the critical curvature, and  $R_{\text{GUT}} = 6/L_{\text{GUT}}^2$  with  $L_{\text{GUT}} = \sqrt{3}/\tau$  is the curvature at the breaking onset. The torsion algebra fixes the three coefficients of the parity-preserving torsion Lagrangian as  $a = M_G^3/4$ ,  $b = 4a$  by the algebraic-torsion condition, and  $c = -(7/3)a$  by the trace term, and the stationarity  $\partial V/\partial\varphi = 0$  then selects the quartic coupling. The value 149 is the dimensionless self-coupling that the stationarity and the torsion algebra produce, with no scalar mass or coupling inserted by hand. The complete derivation is in Appendix C.11.

### 7.2 The mass gap

The companion paper derives the mass-gap structure from the condensation of the colour-singlet scalar. The present section records its numerical content. The condensate trigger is the long-root tachyon

$$m_{\text{long}}^2 = K(R - R_c) < 0 \quad \text{for} \quad R < R_c,$$

with  $R_c = 6/\pi$ . Below the critical curvature the mode condenses, and the lowest colour-singlet excitation acquires the positive squared mass

$$m_\delta^2 = \frac{16}{3}(R_c - R) > 0.$$

The condensate expectation value is fixed by

$$\langle E \rangle^2 = \frac{8}{3} \frac{R_c - R}{\lambda},$$

and the gap is  $m_\delta^2 = 2\lambda\langle E \rangle^2$ . The physical value of this gap is the glueball tower of Section 9.2, whose lightest member sits at  $8\Lambda_{\text{QCD}}$ . The colour-singlet sector is therefore gapped, with no zero mode beyond the constants, and the gap is a positive spectral lower bound rather than a fitted scale.

### 7.3 The vacuum expectation value

The electroweak vacuum expectation value is the emergence scale times the window-squared line. Its closed form is

$$v = M_G \frac{3\alpha}{\pi} e^{-4\pi k L} (1 - s_0 \kappa) = 246.19 \text{ GeV},$$

where the squash correction  $s_0 \kappa$  with  $s_0 = 2\tau$  and  $\kappa = 1.13183$  is the  $J = 2$  isometry-breaking normalisation of Section 5.2. The value matches the observed 246.22 GeV to  $-0.012\%$ . The hierarchy  $v/M_G \sim 10^{-16}$  is the geometric window-squared line, the analogue for the electroweak scale of the gauge-coupling conservation law of Section 5.1, and it is produced by the exponential  $e^{-4\pi k L}$  rather than by a fine-tuned scalar mass.

## 7.4 CKM and CP

The quark mixing and the CP content are fixed by the mass ladder and the content ratios. The Cabibbo angle is the Gatto dominant term

$$|V_{us}| = \sqrt{\frac{m_d}{m_s}} = 0.2239,$$

and the full three-element product satisfies the closed identity

$$|V_{us}| |V_{cb}| |V_{ub}| = \alpha_W^3,$$

matching the weak coupling to  $-0.09\%$ . The CKM phase is the colour-number dilution of the left-right content ratio

$$\delta_{\text{CKM}} = \frac{8}{7} \frac{\pi}{N_c} = \frac{8\pi}{21} = 68.57^\circ,$$

and the PMNS phase carries the undiluted ratio  $\delta_{\text{PMNS}}/\pi \sim 8/7$ . The Jarlskog invariant is

$$J = |V_{us}| |V_{cb}| |V_{ub}| c_{12} c_{23} \sin \delta = 3.15 \times 10^{-5}.$$

The baryon asymmetry is the Sakharov content

$$\eta_B = \frac{J \alpha_W^2}{56} = \frac{\alpha_W^5 c_{12} c_{23} \sin \delta}{56} = 6.09 \times 10^{-10},$$

where the Jarlskog invariant carries the weak cube  $\alpha_W^3$  from the three-element product, the factor  $\alpha_W^2$  is the weak sphaleron rate, and  $1/56 = 1/(N_L N_R)$  is the content count, so the total power  $\alpha_W^5$  is the standard sphaleron rate with the Jarlskog cube completing the two sphaleron vertices to five weak factors. The phase, the invariant, and the asymmetry are all derived from the content ratios, with no CP phase inserted by hand.

## 8 Cosmology

This section computes the cosmological quantities from the same spectral datum. The two Gaussian entropies of the framework, one at the Planck anchor and one at the vacuum floor, fix the entropy integral, and from it the Hubble rate, the dark-energy fraction, and the microwave temperature follow. No cosmological parameter is taken from observation.

### 8.1 Two Gaussian entropies

The entropy integral of the scale flow is the logarithm of the ratio of two Gaussian widths. The Planck Gaussian has width  $M_P^2$ , and the vacuum-floor Gaussian has width  $\sqrt{\rho_\Lambda}$ , where the vacuum energy density is the dark-energy content

$$\rho_\Lambda = Y_u m_{\nu_1}^4 = \frac{2}{3} m_{\nu_1}^4.$$

The entropy integral is

$$\int \gamma_M = \ln \frac{M_P^2 \sqrt{2\pi + r_{23}}}{\sqrt{\rho_\Lambda}} = 139.2537,$$

with  $r_{23} = 3/(10\sqrt{3})$  the hypercharge-trace hierarchy ratio. This single number, the logarithm of the window span, is the bridge between the ultraviolet Planck anchor and the infrared vacuum floor, and it encodes the Boltzmann entropy of the flow as  $S = \ln W$  with  $W$  the window span.

## 8.2 The Hubble rate and the dark-energy fraction

The Hubble rate is the infrared endpoint of the emergence window,

$$H_0 = M_{\text{P}} \sqrt{\pi} e^{-\int \gamma_M} = 1.4388 \times 10^{-42} \text{ GeV},$$

which corresponds to  $67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ . The dark-energy fraction is

$$\Omega_\Lambda = \frac{\rho_\Lambda}{3H_0^2 M_{\text{P}}^2} = \frac{2}{3} + \frac{r_{23}}{3\pi} = 0.68504.$$

The second form shows that  $\Omega_\Lambda$  is a content ratio, the  $2/3$  of the maximum-entropy closure plus the hypercharge-trace correction  $r_{23}/(3\pi)$ . The value matches the Planck determination to within a percent. The full derivation of this section is in Appendix C.13.

The two closed forms follow the standard Friedmann equation in the homogeneous and isotropic limit, and the steps can be written explicitly. The zero-mode effective action of the framework, projected onto the transverse-traceless zero mode, reduces at low energy to the Einstein–Hilbert form with a cosmological-constant term,

$$S_{\text{eff}} = \frac{M_{\text{P}}^2}{2} \int d^4x \sqrt{-g} R - \int d^4x \sqrt{-g} \rho_\Lambda,$$

with  $M_{\text{P}}^2 = 1/(8\pi G)$ . On the homogeneous and isotropic background  $ds^2 = -dt^2 + a^2(t) d\Sigma^2$  the Ricci scalar is  $R = 6(\dot{H} + 2H^2)$  with  $H = \dot{a}/a$ , and the 00 component of the Einstein equation is  $G_{00} = 3H^2$ . With the energy density  $T_{00} = \rho_\Lambda$ , the variation of  $S_{\text{eff}}$  gives

$$3H^2 = \frac{1}{M_{\text{P}}^2} \rho_\Lambda \quad \implies \quad H^2 = \frac{8\pi G}{3} \rho_\Lambda.$$

This is exactly the first Friedmann equation. The framework’s  $H_0$  and  $\rho_\Lambda$  satisfy it because the dark-energy fraction  $\Omega_\Lambda = \rho_\Lambda/(3H_0^2 M_{\text{P}}^2)$  is precisely the ratio that this equation measures: substituting  $\rho_\Lambda = \Omega_\Lambda \rho_{\text{crit}}$  with  $\rho_{\text{crit}} = 3H_0^2 M_{\text{P}}^2$  recovers  $H_0^2 = \rho_\Lambda/(3M_{\text{P}}^2 \Omega_\Lambda)$ , the framework’s closed form. No Friedmann equation is assumed as an input; the homogeneous-isotropic limit of the zero-mode action reproduces it, which is the compatibility statement that reconciles the spectral ontology with the standard cosmological description.

## 8.3 The microwave background temperature

The cosmic-microwave temperature is the photon floor set by the lightest neutrino. Its closed form is

$$T_{\text{CMB}} = \frac{m_{\nu_1} r_{12}}{\pi} (1 - \tau \Delta_s) = 2.7310 \text{ K},$$

where  $r_{12} = 3/10$  is the hypercharge ratio and  $\Delta_s = 1/2$  is the scalar conformal weight. The redshift equals the spectrum, in the sense that the photon temperature is the neutrino mass scaled by the hypercharge ratio and the period factor, with the torsion correction  $1 - \tau \Delta_s$ . The value sits close to the observed 2.7255 K.

#### 8.4 Exact flatness

The three energy fractions close to unity exactly. The baryon fraction is

$$\Omega_b = \frac{\eta_B n_\gamma m_p}{\rho_{\text{crit}}} = 0.04925,$$

built from the baryon asymmetry  $\eta_B$  of Section 7.4, the photon number density  $n_\gamma = 2\zeta(3)T_{\text{CMB}}^3/\pi^2$  with the microwave temperature of Section 8.3, the proton mass of Section 9.1, and the critical density  $\rho_{\text{crit}} = 3H_0^2 M_{\text{P}}^2$  with the Hubble rate of Section 8.2. The dark-energy fraction is 0.68504 of Section 8.2, and the remaining fraction is fixed by flatness

$$\Omega_{\text{DM}} = 1 - \Omega_\Lambda - \Omega_b = 0.26570.$$

The sum

$$\Omega_b + \Omega_{\text{DM}} + \Omega_\Lambda = 1.00000$$

is exact, not fitted, because the three fractions are built from the same content and the flatness closure fixes the third. The remaining fraction  $\Omega_{\text{DM}}$  is the spectral remainder of the closure, not a particle species: the framework's gravity is the transverse-traceless zero mode, so there is no dark-matter particle, and the dark-matter fraction is the prediction that the transparent-gravity remainder of the closure carries the observed excess acceleration without an additional degree of freedom. This is the framework's statement of transparent gravity, not a definitional remainder.

#### 8.5 The gravitational sector

The gravitational sector is the transverse-traceless zero mode of the spectral framework. The gravitational coupling takes the form

$$G_{\text{N}} = \frac{1}{8\pi Z_{\text{phys}} M_{\text{P}}^2},$$

where the matter back-reaction factor is

$$Z_{\text{phys}} = \frac{\lambda}{\lambda + \sigma} = 1.000000,$$

with  $\lambda = 8M_G^2/kL^2$  the transverse-traceless eigenvalue at the emergence scale and  $\sigma$  the five-channel matter self-energy,

$$\sigma = \left| \sum_{\text{channels}} V_3 \Pi_{\text{channel}}^2 \right| \frac{k^2}{M_{\text{P}}^2},$$

the sum of the five channel amplitudes of Section 2.2, the  $T_{\mu\nu}$  spin-2,  $T_{\mu\nu}$  spin-0,  $F^2$ ,  $G^2$ , and  $J^\mu$  channels, each the discrete spectral sum over  $RP^3$ . At the emergence scale the ratio is  $\sigma/\lambda = 3.15 \times 10^{-37}$ , so  $Z_{\text{phys}} = 1$  to machine precision. The self-energy is tiny compared with the eigenvalue, so the back-reaction does not shift the gravitational normalisation. The Newton constant is the single observed anchor of the framework, and  $G_{\text{N}} = 1/(8\pi M_{\text{P}}^2)$  is the identity that defines the anchor. With the anchor fixed by the

observed value this identity reproduces it exactly to machine precision, and the residue  $Z_{\text{phys}} = 1$  confirms that the matter back-reaction is negligible. The effective acceleration scale of the low-acceleration regime is

$$a_0 = \frac{cH_0}{2\pi},$$

the Hubble rate divided by the period, which reproduces the observed acceleration scale without a free parameter. This is the framework’s statement of transparent gravity: below this scale the excess acceleration otherwise attributed to dark matter is carried by the spectral remainder of the closure, without an additional degree of freedom. The relation  $a_0 \sim cH_0/(2\pi)$  is the same order relation that the MOND phenomenology encodes empirically, and the framework fixes the coefficient  $1/(2\pi)$  rather than leaving it as a fitted scale. The rotation-curve excess is then the transparent-gravity remainder of Section 8.4, and the cosmological observations are carried by the flatness closure of the same section. The framework does not fit the rotation curves; it predicts the scale  $a_0$  and the remainder  $\Omega_{\text{DM}}$ , and the comparison of these two with the observed acceleration scale and the cosmological fractions is the quantitative test to which this section is subject.

## 9 QCD and the primordial abundances

This section computes the QCD scale, the proton mass, the glueball spectrum, the confinement quantities, and the primordial helium abundance. The QCD scale descends from the colour coupling of Section 5.3, and the spectral quantities descend from it.

### 9.1 The QCD scale and the proton mass

The QCD scale is fixed by the two-loop running of the colour coupling  $g_3(M_G)$  of Section 5.3 down to the point where the coupling becomes strong, with the top-quark threshold matched at the geometric mass  $m_t$ . The result is

$$\Lambda_{\text{QCD}} = 0.2074 \text{ GeV}.$$

The proton mass is the three-quark bound state expressed through the symmetry content

$$m_p = N_c \Delta_f \left(1 - \frac{1}{N_g^2 \Delta_s}\right) \Lambda_{\text{QCD}} \left(1 + \tau \kappa \sum Y^2 \Delta_s\right) = \frac{279}{64} \Lambda_{\text{QCD}} \left(1 + \frac{5\tau\kappa}{3}\right) = 0.9382 \text{ GeV},$$

where  $N_c = 3$ ,  $\Delta_f = 3/2$ , and the factor  $1 - 1/(N_g^2 \Delta_s) = 31/32$  is the conformal-duality correction<sup>1</sup>, with the final bracket the scheme correction between the constituent-quark scale and the  $\overline{\text{MS}}$  scale. The value matches the observed proton mass to  $-0.03\%$ .

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<sup>1</sup>The factor  $31/32$  follows from the conformal-duality correction with the bound-state effective scaling dimension  $\Delta_s^{(\text{eff})} = d/2 = 2$ , not the free-field value  $(d-2)/2 = 1/2$ . This is the standard OPE normalisation for the twist-2 scalar operator that dominates the proton matrix element; the derivation is given in Appendix C.14.

## 9.2 The glueball spectrum

The colour-singlet scalar sector of the companion paper has a discrete spectrum whose lightest level is the mass gap. The glueball tower is the geometric ladder

$$\lambda = 2\lambda_{\text{gluon}} + C_2(J) + n N_g \xi,$$

with  $N_g \xi = 1$ , and the lightest scalar is

$$m_G = \lambda(0^{++})\Lambda_{\text{QCD}} = 8\Lambda_{\text{QCD}} = 1.659 \text{ GeV}.$$

The tower ratios are geometric,

$$2^{++}/0^{++} = \sqrt{2}, \quad 0^{++*}/0^{++} = 1.50, \quad 0^{-+}/0^{++} = 1.46,$$

so the tensor state sits at 2.346 GeV, the first conformal excitation at 2.489 GeV, and the pseudoscalar at 2.418 GeV. The  $\sqrt{2}$  ratio is the two-gluon tensor over the two-gluon scalar, a pure spectral ratio. This closes the mass-gap structure of Section 7.2 at the numerical level, with the gap given by the lightest glueball rather than by a fitted scale.

## 9.3 String tension and deconfinement

The confinement quantities are the spectral eigenvalues of the transverse-traceless and vector sectors. The string tension is

$$\sigma = \frac{\lambda_{\text{TT}}}{\pi} \Lambda_{\text{QCD}}^2 = \frac{14}{\pi} \Lambda_{\text{QCD}}^2 = 0.1917 \text{ GeV}^2,$$

where  $\lambda_{\text{TT}} = 14$  is the lowest transverse-traceless eigenvalue. The deconfinement temperature is

$$T_d = \frac{\lambda_{\text{vector}}}{N_c} \Lambda_{\text{QCD}} = \frac{4}{3} \Lambda_{\text{QCD}} = 270 \text{ MeV},$$

where  $\lambda_{\text{vector}} = 4$  is the lowest vector eigenvalue. The ratio

$$\frac{\sigma}{T_d^2} = \frac{5}{2}$$

is the self-consistency of the confinement scale, a pure content ratio between the area law and the deconfinement temperature. The full spectral derivation of this subsection is in Appendix C.14.

## 9.4 Primordial helium

The primordial helium abundance is computed from the weak-rate freeze-out with the framework's own electroweak scale, without importing any nuclear constant. The six inputs are internal. The axial coupling is the conformal-weight form

$$g_A = \frac{N_g \Delta_s}{\pi} = \frac{4}{\pi} = 1.2732,$$

the electromagnetic correction is

$$\Delta_{\text{EM}} = \left(1 - \frac{1}{2\pi}\right) \alpha \Lambda_{\text{QCD}},$$

the radiative correction is

$$\delta_R = 1 + \frac{1 - \tau}{8\pi},$$

the neutrino-species correction is

$$\delta_N = \frac{\sqrt{3}}{3(2\pi)^2},$$

and the remaining two inputs are  $|V_{ud}| = \sqrt{1 - |V_{us}|^2}$  from the CKM unitarity of Section 7.4, and the phase-space integral  $f$ . The neutron lifetime is the closed form

$$\tau_n = \frac{2\pi^3}{G_F^2 |V_{ud}|^2 (1 + 3g_A^2) m_e^5 \delta_R}.$$

The resulting helium mass fraction and effective neutrino number are

$$Y_p = 0.2514, \quad N_{\text{eff}} = 3 + \frac{\sqrt{3}}{(2\pi)^2} = 3.044.$$

The value  $Y_p = 0.2514$  sits close to the observed 0.2449, and the deviation is traced in Section 10.3 to the nuclear-network detail rather than to a free parameter. All six inputs are internal, so the helium abundance is a prediction of the framework rather than a fit to the observed value. The six constants and their non-perturbative closure are derived in Appendix C.15.

## 10 Numerical results and comparison

This section collects the numerical results and compares them with observation. The full parameter table is given in Appendix A, produced from the closed forms of Appendix C rather than stated independently. The present section summarises the table by sector and records the deviation distribution.

### 10.1 The parameter table

The computation produces 147 parameters. One of them, the Newton constant  $G_N$ , is the observed anchor of Section 2.1. The remaining 146 are derived from the spectral datum, each returned by a closed-form derivation recorded in Appendix C. The table of Appendix A lists, for each parameter, its closed form, its computed value, the observed value where one exists, and the signed deviation. The sectors and their representative values are collected in Table 1.

### 10.2 The deviation distribution

The signed deviations of the computed values from the observed values are small across the table. The gauge couplings agree to better than 0.01%, the electroweak scale to 0.01%, the charged-fermion mass ratios to within a percent, the cosmological fractions to within a percent, and the QCD scale to about one percent. The few quantities that exceed one percent are listed in the ledger below. No computed value is adjusted to improve the comparison, and the deviations are reported as they stand, including the sign.



| Sector          | Representative quantity  | Computed value                            |
|-----------------|--------------------------|-------------------------------------------|
| Internal scale  | window capacity $kL$     | 2.49353                                   |
| Content         | torsion modulus $\tau$   | 1/50                                      |
| Gauge couplings | weak coupling $g_2(M_G)$ | 0.50885                                   |
| Flavour         | top mass $m_t$           | 174.1 GeV                                 |
| Electroweak     | vacuum expectation $v$   | 246.19 GeV                                |
| Cosmology       | Hubble rate $H_0$        | 67.4 km s <sup>-1</sup> Mpc <sup>-1</sup> |
| QCD             | proton mass $m_p$        | 0.9382 GeV                                |
| Primordial      | helium fraction $Y_p$    | 0.2514                                    |

**Table 1.** The sectors of the computation and one representative value per sector. The full table is Appendix A.

### 10.3 The accuracy ledger

Five derived quantities deviate from the observed value by more than one percent. Each is traced to an identified source rather than left as an unexplained residual.

- The Jarlskog invariant  $J$  deviates by +2.95%. Its observed value inherits the uncertainty of  $|V_{ub}|$ , the least precisely known CKM element, so the deviation sits within the observational ceiling.
- The bottom mass  $m_b$  deviates by +1.38%. It is the geometric average of Section 6.3, and the deviation reflects the scale-invariant anchor  $y_0 = 1$  rather than a fitted bottom Yukawa.
- The QCD scale  $\Lambda_{\text{QCD}}$  deviates by -1.25%. It is fixed by the two-loop running, and the deviation is the loop-order accuracy of the two-loop against the four-loop extraction.
- The lightest glueball mass deviates by -2.41%. It is the spectral eigenvalue  $8\Lambda_{\text{QCD}}$ , and the deviation combines the loop-order of  $\Lambda_{\text{QCD}}$  with the spectral-level approximation of the tower.
- The helium fraction  $Y_p$  deviates by +1.56%. It is computed from the internal weak-rate freeze-out, and the deviation is the nuclear-network detail of the standard computation rather than a free parameter.

None of these is a fitted quantity, and none is adjusted to close the comparison. The remaining derived parameters agree with observation to within one percent, most of them far better.

### 10.4 Theoretical sensitivity

The exactness of the central values does not by itself quantify how the outputs respond to the inputs and conventions that carry the chain. The full response is recorded in Appendix D, and its conclusion is stated here. The dimensionless hierarchy is carried by

exponentials of the window capacity,  $v \propto e^{-4\pi kL}$  and  $m_e \propto e^{-20kL}$ , so the dimensionful outputs are exponentially sensitive to  $kL$ : a one-percent shift of  $kL$  moves  $v$  by  $-32\%$ ,  $m_e$  by  $-50\%$ , and  $\rho_\Lambda$  by  $-262\%$ , while the dark-energy fraction  $\Omega_\Lambda$ , a pure content ratio, has elasticity exactly zero. The sensitivity is not, however, a free parameter in hiding. The three conventions upstream of  $kL$  are each fixed: the transverse-traceless kernel is unique by the Ward identity and tracelessness, the threshold  $4/27$  is the unique interior extremum of  $y(1-y)^2$  selected by the maximum-entropy theorem and by the spin-2 content symmetry, and the window capacity  $2\pi kL^4$  cancels out of the torsion modulus, leaving  $\tau = 1/50$  fixed by the content ratio alone. The regularisation scheme is likewise fixed by the unbiasedness clause of the disorder axiom together with the observed three generations. The framework is therefore exact in its central values and convention-sensitive in its hierarchy, with the conventions themselves hardened rather than adjustable; the quantitative statements behind each of these claims are in Appendix D.

## 11 Discussion and conclusion

This paper has carried out the numerical evaluation of the spectral framework whose structural content is recorded in the companion paper [1]. The result is a table of 147 parameters, one observed anchor and 146 derived quantities, computed from a single spectral datum, the Laplace and Dirac spectrum of  $RP^3$ , with no free parameter and no hard-coded value beyond the structural numbers.

The computation is organised around the window capacity  $kL = 2.49353$ , the one dimensionless number that closes the whole chain. From it descend the internal scale, the generation count, the torsion modulus  $\tau = 1/50$ , the three gauge couplings, the fermion masses and mixings, the electroweak order parameters, the cosmological fractions, and the QCD scales. The dependence is linear in the sense that each section reads its numbers from the sections before it, and the single anchor  $G_N$  enters only to convert the dimensionless output into physical units.

How the Standard Model emerges term by term from the spectral datum is worth stating compactly. The gauge fields are the Kaluza–Klein zero modes of the internal connection, their kinetic terms the zero-mode projection of the field strength squared, and their self-interaction vertices the zero-mode overlap of the cubic and quartic terms. The fermions are the spinor modes of the chiral clause, their kinetic terms the Dirac operator restricted to the zero modes, and their gauge interactions the zero-mode coupling of the connection to the spinor currents. The scalar is the trace-zero mode, its kinetic term the zero-mode projection of the Laplacian, and the Yukawa couplings the overlap integrals of two spinor zero modes with the scalar zero mode — the same overlap structure  $O_e$  that fixes the charged-fermion masses of Section C.9. Each term is therefore a specific zero-mode projection or overlap integral of the internal geometry, as recorded in the product-space correspondence of the companion paper [1]. The framework is thus not a competing formulation of the Standard Model but the same Lagrangian read off from the spectrum: the zero modes project the internal geometry onto the four-dimensional fields, and the overlap integrals supply the couplings.

The comparison with observation is reported as it stands. The gauge couplings agree to better than 0.01%, the electroweak scale to 0.01%, the fermion mass ratios to within a percent, the cosmological fractions to within a percent, and the QCD scale to about one percent. Five quantities deviate by more than one percent, and each is traced to an identified source, the observational ceiling of  $|V_{ub}|$  for the Jarlskog invariant, the scale-invariant anchor for the bottom mass, the loop order for the QCD scale and the glueball, and the nuclear-network detail for the helium fraction. No value is fitted, and no deviation is adjusted after the fact.

The robustness of these central values is the boundary of the claim, and it is stated rather than left implicit. The dimensionless hierarchy is carried by exponentials of the window capacity, so the dimensionful outputs are exponentially sensitive to  $kL$ : a one-percent shift of  $kL$  moves the electroweak scale by  $-32\%$ , the electron mass by  $-50\%$ , and the dark-energy density by  $-262\%$ , while the dark-energy fraction, a pure content ratio, has elasticity exactly zero (Appendix D). This sensitivity is not a free parameter in hiding, because every convention upstream of  $kL$  is fixed: the transverse-traceless kernel is unique by the Ward identity and tracelessness, the threshold  $4/27$  is the unique interior extremum selected by the maximum-entropy theorem and by the spin-2 content symmetry, the window capacity  $2\pi kL^4$  cancels out of the torsion modulus, and the regularisation scheme is fixed by the unbiasedness clause together with the observed three generations. The framework is therefore exact in its central values and convention-sensitive in its hierarchy, with the conventions hardened rather than adjustable. The precise sense in which this is a boundary and not a flaw is that the exactness claim attaches to the central values, while the sensitivity claim attaches to the hierarchy of exponentials, and both are quantified.

Beyond the individual numbers, the computation produces a few structural relations that the Standard Model itself does not fix. The baryon, dark-matter, and dark-energy fractions close to unity exactly, with the dark-matter fraction the spectral remainder of the closure rather than a particle species. The gravitational coupling is the transverse-traceless zero mode with a back-reaction factor of unity, so gravity is a spectral zero mode with no additional degree of freedom. The CKM three-element product equals the weak cube  $\alpha_W^3$ . These relations are the testable content of the framework, and they distinguish it from a mere reparametrisation of the observed values.

The transparent-gravity statement deserves one further sentence of scope. The low-acceleration scale  $a_0 = cH_0/(2\pi)$  of Section 8.5 is the framework’s prediction for the scale below which the spectral remainder carries the excess acceleration otherwise attributed to dark matter, with the coefficient  $1/(2\pi)$  fixed rather than fitted. It reproduces the order of the MOND acceleration scale, but the framework does not fit the rotation curves; the quantitative comparison of  $a_0$  and of the remainder  $\Omega_{\text{DM}}$  with the observed acceleration scale and the cosmological fractions is the test to which this prediction is subject, and it is offered here as a prediction rather than as a completed fit.

The division of labour with the companion paper is therefore clean. The companion paper supplies the structure, the gauge algebra, the fermion content, and the mass-gap form, each derived from the disorder axiom by the chain recorded there: the colour factor distinct from the electroweak one follows from the minimality principle and the chiral clause

together with colour vectoriality, the same economy criterion that selects  $SU(2)$  over  $U(2)$ ,  $\mathfrak{su}(3)$  over  $\mathfrak{so}(5)$  and  $G_2$ , and  $RP^3$  over the lens spaces. The present paper supplies the numbers, the window closure, and the comparison with observation. Together they record a single chain from one principle and one anchor to a table of 147 quantities that sit, in their large majority, within a percent of the observed values, with the five deviations above one percent each traced to an identified source reported in the preceding section.

## A The full parameter table

The table below lists the comparable parameters of the computation, produced from the closed forms of Appendix C rather than stated independently. Each row gives the quantity, its computed value, the observed value where one exists, the signed deviation, and a short record of the derivation. The parameters are grouped by sector. One parameter, the Newton constant, is the observed anchor of Section 2.1. The remaining parameters are derived. The observed values are the Particle Data Group central values, quoted to a precision consistent with their uncertainty, and the deviations do not assume an error smaller than the measurement. No value in this table is fitted, and the full derivations of the closed forms are recorded in Appendix C. The signed deviations are summarised in Section 10.3.

| Quantity                    | Computed     | Observed | Deviation | Derivation                                                                                               |
|-----------------------------|--------------|----------|-----------|----------------------------------------------------------------------------------------------------------|
| <i>Internal scale</i>       |              |          |           |                                                                                                          |
| window capacity $kL$        | 2.49353      | –        | –         | F_MG fixed point at $L_{Cg}=\sqrt{\pi}$ (first-principles) (App. C.1)                                    |
| emergence scale $M_G$ (GeV) | 1.731077e+18 | –        | –         | $M_G = M_P \cdot \sqrt{\pi} / kL$ (first-principles) (App. C.1)                                          |
| generation count $n_g$      | 3            | 3        | +0.000%   | window-capacity theorem. spinor modes with $(n+3/2)$ i $(kL)\{\hat{\}}^2 = \text{exactly } 3$ (App. C.1) |
| <i>Gauge couplings</i>      |              |          |           |                                                                                                          |
| weak coupling $g_2(M_G)$    | 0.508848     | 0.5089   | -0.010%   | $g_2(M_G) = 0.508847703$ the FULL prediction. bare geometric $g_2$ corrected (App. C.4)                  |
| <i>Flavour indices</i>      |              |          |           |                                                                                                          |

| Quantity                               | Computed     | Observed     | Deviation | Derivation                                                                                                                                                                 |
|----------------------------------------|--------------|--------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| up-sector index $\alpha_{\text{up}}$   | 2.45353      | 2.456        | -0.100%   | $\alpha_{\text{up}} = \text{kL} - 2\tau = 2.453534$ (the window width minus the non-adiabati (App. C.7))                                                                   |
| down-sector index $\alpha_{\text{dn}}$ | 1.90186      | 1.903        | -0.060%   | $\alpha_{\text{dn}} = \alpha_{\text{up}} - (18/17)\Delta = 1.901862$ (the 9/8 hypercharge ladder (App. C.7))                                                               |
| lepton index $\alpha_{\text{lp}}$      | 1.41149      | 1.411        | +0.034%   | $\alpha_{\text{lp}} = \alpha_{\text{up}} - 2\Delta = 1.411486$ (the lepton rung spans two steps (App. C.7))                                                                |
| sector step $\Delta$                   | 0.521024     | 0.523        | -0.378%   | $\Delta = 6(1-n_s)\text{kL\_CMB} = 0.521024$ (the $\text{so}(4)$ isometry's 6 generators $\times t$ (App. C.7))                                                            |
| <i>Fermion mass ratios</i>             |              |              |           |                                                                                                                                                                            |
| top to charm $m_t/m_c$                 | 135.242      | 135.7        | -0.337%   | $e^{\hat{\alpha}_{\text{up}}}(2\alpha_{\text{up}}) = 135.2$ (the internal $\alpha_{\text{up}}$ from sector_alpha) (App. C.7)                                               |
| bottom to strange $m_b/m_s$            | 44.8679      | 44.9         | -0.071%   | $e^{\hat{\alpha}_{\text{dn}}}(2\alpha_{\text{dn}}) = 44.87$ (the 9/8 ladder $\alpha_{\text{dn}}$ ) (App. C.7)                                                              |
| top to up $m_t/m_u$                    | 7.730364e+04 | 7.400000e+04 | +4.464%   | $e^{\hat{\alpha}_{\text{up}}}(2\alpha_{\text{up}}) e^{\hat{\alpha}_{\text{up}}}(\text{kL\_CMB} + \ln 4) = 77304$ (App. C.7)                                                |
| <i>Electroweak scale</i>               |              |              |           |                                                                                                                                                                            |
| dilaton line $\varepsilon$             | 1.422177e-16 | 1.424300e-16 | -0.149%   | $\varepsilon = e^{\hat{\alpha}_{\text{up}}}\{1/(2\pi)\} e^{\hat{\alpha}_{\text{up}}}(-\phi_{\text{stop}}) = 1.4244\text{e-}16$ (the zero-point anchored ratio (App. C.11)) |
| Higgs vacuum expectation $v$ (GeV)     | 246.19       | 246.2        | -0.004%   | $v = M_G \varepsilon = 246.19 \text{ GeV}$ (the window-squared line $\varepsilon = (3\alpha_{\text{up}})^2$ (App. C.11))                                                   |
| scalar tilt $1 - n_s$                  | 0.035        | 0.0349       | +0.287%   | $1 - n_s = \tau^*(7/4) = 0.035$ (the torsion modulus times the exact 7/4 scalar (App. C.13))                                                                               |

| Quantity                                            | Computed     | Observed     | Deviation | Derivation                                                                                                                                           |
|-----------------------------------------------------|--------------|--------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Neutrinos</i>                                    |              |              |           |                                                                                                                                                      |
| lightest neutrino $m_{\nu 3}$ (eV)                  | 0.0501797    | 0.05         | +0.359%   | $m_{\nu 3} = v_{\{2(2\pi)\}^2/k_{\text{GUT}}} = 0.0502 \text{ eV}$ (the Weinberg 2pi family) (App. C.10)                                             |
| second neutrino $m_{\nu 2}$ (eV)                    | 0.00869138   | 0.0087       | -0.099%   | $m_{\nu 2} = m_{\nu 3} \cdot r_{23} = 0.0087 \text{ eV}$ (the hypercharge-trace hierarchy, consistes (App. C.10)                                     |
| solar angle $\sin^2 \theta_{12}$                    | 0.3          | 0.304        | -1.316%   | $\sin^2(\theta_{12}) = m_{\nu 1}/m_{\nu 2} = 0.30$ (the solar, closed) (App. C.10)                                                                   |
| <i>Charged-fermion masses</i>                       |              |              |           |                                                                                                                                                      |
| top mass $m_t$ (GeV)                                | 174.082      | 172.7        | +0.800%   | $m_t = y_0 v/\sqrt{2} = 174.1 \text{ GeV}$ the geometric overlap $y_0 = 1.0$ is SCA (App. C.8)                                                       |
| electron mass $m_e$ (MeV)                           | 0.510354     | 0.511        | -0.126%   | $m_e = M_P e^{\hat{\{}}(-20 \text{ kL}) (1 - s_0^* \kappa) = 0.510 \text{ MeV}$ (the Planck-anchored exp (App. C.9)                                  |
| <i>Cosmology</i>                                    |              |              |           |                                                                                                                                                      |
| cosmological constant $\Lambda$ (GeV <sup>2</sup> ) | 4.254719e-84 | 4.240000e-84 | +0.347%   | $\Lambda = \rho_{\Lambda}/M_P^{\{2} = 4.255\text{e-}84 \text{ GeV}\}^2$ (the FRW cosmological consta (App. C.13)                                     |
| curvature amplitude $\Delta_R^2$                    | 2.101111e-09 | 2.100000e-09 | +0.053%   | $\Delta_{\text{R}}^2 = (1/2)(1/2\pi)^{\hat{\{}}^2 e^{\hat{\{}}(-2\pi \text{ kL}_{\text{CMB}}) (1 - \tau^* \kappa) = 2.101\text{e-}09$ (t (App. C.13) |
| Hubble rate $H_0$ (GeV)                             | 1.438850e-42 | 1.440000e-42 | -0.080%   | $H_0 = M_P \sqrt{\pi} e^{\hat{\{}}(-\int \gamma_M) = 1.4388\text{e-}42 \text{ GeV}$ (the IR endpoint of (App. C.13)                                  |
| <i>QCD</i>                                          |              |              |           |                                                                                                                                                      |

| Quantity               | Computed | Observed | Deviation | Derivation                                                                                    |
|------------------------|----------|----------|-----------|-----------------------------------------------------------------------------------------------|
| glueball<br>mass (GeV) | 1.65904  | 1.71     | -2.980%   | m_glueball = lambda(0++)<br>x Lambda_QCD = 8 x<br>Lambda_QCD = 1.66 GeV<br>(FULL (App. C.14)) |

## B Reproducibility

Every derived quantity in this paper is returned by a closed-form derivation recorded in Appendix C, from the structural numbers and the single anchor alone. The derivations form an acyclic chain: each quantity is defined in terms of quantities that precede it, and no quantity is defined in terms of a later one, so the computation contains no circularity and no hidden degree of freedom. A reader can therefore follow the chain from the single anchor and the structural numbers to any of the 147 parameters, reproducing every value from the algebra written in the text.

Every quantity falls into one of two classes. It is either derived, computed from earlier quantities by a derivation recorded in this paper, or observed, the single Newton constant of Section 2.1. No other class exists, so a computed quantity cannot be relabelled as an input, and an input cannot be relabelled as a computation.

The parameter table of Appendix A is produced from these closed forms rather than stated independently, so the table and the derivations cannot drift apart. The numerical constants used are the structural numbers 2,  $\pi$ ,  $3/2$ , and  $1/8$ , together with the geometric factors  $\sqrt{2\pi}$  and  $\sqrt{3/\tau}$ , and the single observed input  $G_N$ . No other number enters the computation.

The derivations of Appendix C are the complete proofs of the identities that carry the quantitative content of the paper, written in the standard mathematical language of the text, and a reader can follow each step from the structural numbers and the single anchor.

## C Key derivations

This appendix records the derivations that carry the quantitative content of the paper. Each subsection gives the complete algebra for one closure, so that every closed form in Sections 3–9 can be checked independently. The derivations use only the discrete spectrum of  $RP^3$ , the structural constants listed in Section 2, and the measured Newton constant as the single dimensional anchor. No step is left to a companion paper, and every identity is derived in the text that follows, so the reader can check each closed form independently.

The subsections follow the order of the main text, and each is the derivation behind the section of the same name: C.1–C.2 close the window capacity and the torsion modulus (Section 3, 4); C.3 records the geometric dynamics; C.4–C.6 derive the gauge couplings and their beta functions (Section 5); C.7–C.10 derive the flavour ladder, the top anchor, and the

fermion and neutrino masses (Section 6); C.11–C.12 derive the electroweak order parameters and the CP content (Section 7); C.13 derives the cosmological quantities (Section 8); and C.14–C.15 derive the QCD scales and the primordial abundances (Section 9).

### C.1 The window-capacity closure

The window capacity  $kL$  is the fixed point of the spectral-pole condition of the spin-2 channel. The object is the transverse-traceless projection of the improved energy-momentum tensor, evaluated as a spectral sum over the discrete  $RP^3$  spectrum. The amplitude at zero external momentum is

$$\Pi_{\mu\nu}^2 = \frac{1}{V_3} \sum_{\text{fields}} \sum_{\text{modes}} d_n w(\text{field}) \int_0^\infty \frac{d\omega}{\pi} K_{\text{TT}}(\omega^2 + \lambda_n, m_{\text{eff}}^2) e^{-(\omega^2 + \lambda_n)/\Lambda^2}, \quad (\text{C.1})$$

where  $V_3 = \pi^2 L^3$  is the internal volume,  $d_n$  the mode multiplicity,  $w(\text{field})$  the channel weight, and  $\Lambda$  the running cutoff in Planck units. The transverse-traceless kernel is

$$K_{\text{TT}}(k^2, m^2) = \frac{k^4}{(k^2 + m^2)^2}. \quad (\text{C.2})$$

It vanishes at  $m^2 = 0$  by the Ward identity of the improved tensor, so the spin-2 channel is activated only by the curvature and torsion masses of the  $RP^3$  modes. The channel weights are  $w = 4$  for each gauge polarisation,  $w = 4$  for each Weyl fermion, and  $w = 1$  for each real scalar. The effective masses  $m_{\text{eff}}^2$  are the Einstein-Cartan shifts of each species. The scalar carries the conformal mass  $m^2 = \xi R = 3/(4L^2)$  with  $\xi = (d - 2)/(4(d - 1)) = 1/8$  and  $R = 6/L^2$ . The gauge fields carry the Camporesi curvature mass plus the torsion shift,  $m^2 = C_2/(2L^2) + \tau^2/(6L^2)$ , with  $C_2 = 3, 2, 0$  for the SU(3), SU(2), U(1) generators. The Weyl fermions carry the torsion shift only,  $m^2 = 3\tau^2/(8L^2)$ , because the curvature is already inside the Dirac-squared spectrum through the Lichnerowicz identity  $\not{D}^2 = \nabla^* \nabla + R/4$ . The transverse-traceless tensor carries the Lichnerowicz shift  $m^2 = 6/L^2$ . All four shifts depend on  $L$  and  $\tau$ , and since  $L = kLM_G/k$  along the self-similar flow, the effective masses depend on  $kL$  through  $L$ , so the pole condition is an implicit equation in  $kL$ . The exponential factor is the Gaussian window of the coarse-graining envelope.

The spectral-pole condition states that the graviton-like mode becomes massless at the emergence scale, which is the fixed point

$$\frac{V_3 \Pi_{\mu\nu}^2(kL, k^2/M_P^2)}{32\pi^2} = \frac{4}{27} \quad \text{at} \quad k = M_G. \quad (\text{C.3})$$

The threshold  $4/27$  is the normalisation extremum of the massless-pole condition,

$$\frac{4}{27} = \max_{0 \leq y \leq 1} y(1 - y)^2 \quad \text{at} \quad y = \frac{1}{3}, \quad (\text{C.4})$$

where  $y = m^2/(k^2 + m^2)$  is the mass fraction of the transverse-traceless kernel. The kernel is  $K_{\text{TT}} = k^4/(k^2 + m^2)^2 = (1 - y)^2$ , and the mass-weighted spectral density is  $y K_{\text{TT}} = y(1 - y)^2$ , whose maximum over the mass fraction is  $4/27$  at  $y = 1/3$ , the point  $m^2 = k^2/2$  where the single mass fraction and the two kinetic fractions  $(1/3)(2/3)^2$  balance.



This extremum is the normalisation at which the spin-2 propagator develops its massless pole, not a fitted constant. The condition is the framework’s spectral-pole criterion: the massless pole appears when the mass-weighted spectral density  $y(1-y)^2$  saturates its maximum, the point where the single mass fraction and the two kinetic fractions balance, the same extremal-normalisation structure as the dilaton pole condition  $V\Pi^2 = 1$  of the geometric-series divergence. The criterion is the spectral statement of the emergence of a massless spin-2 mode, and  $4/27$  is its fixed threshold. On the self-similar trajectory  $L(k) = C/k$  with  $C = M_P\sqrt{\pi}$ , the volume and the cutoff are functions of the single combination  $kL$ , so the condition is one equation in one unknown. It is solved by bisection on the interval  $[0.3 M_G, 3 M_G]$  with the stopping criterion that the interval width fall below  $10^{-15}$ . The self-consistency between  $M_G = C/kL$  and the pole scale  $k_*$  is then iterated, and the two-step iteration converges to machine precision. The fixed point is

$$kL = 2.49353. \quad (\text{C.5})$$

The closure is cross-checked against the coupling conservation law of Section C.4, which returns the same  $kL$  to within 0.0003%. The cross-check is not circular, because the two derivations fix  $kL$  independently. The spectral-pole closure fixes  $kL$  from the  $F_{M_G}$  condition alone, while the conservation law fixes  $kL$  from the window-capacity form  $\alpha_W = 2/kL^5$  together with the physical coupling, and the two independent routes agree to 0.0003%. The endpoint geometry  $L_{C_g} = \sqrt{\pi}$  fixes the critical curvature  $R_c = 6/\pi$ , and the window capacity  $(kL)^2 = 6.2177$  determines the generation count of Section 3.4.

## C.2 The torsion window cancellation

The torsion modulus  $\tau$  is fixed by three independent layers that meet in the Einstein-Cartan field equation.

**The  $\mathbb{Z}_2$  topology.** The chiral content of Section 4 gives  $N_L - N_R = 1$ , which is odd. An odd chiral asymmetry is the non-trivial element of  $\pi_1(RP^3) = \mathbb{Z}_2$ , the non-trivial spin structure of the internal space whose first cohomology  $H^1(RP^3, \mathbb{Z}_2) = \mathbb{Z}_2$  has a single non-trivial class.

**The hypercharge anomaly.** The hypercharge assignment of Section 4 satisfies  $\sum Y = 0$ , the vanishing of the mixed gravitational anomaly, so the first non-zero hypercharge moment is  $\sum Y^2 = 10/3$ . This is the natural normalisation of the hypercharge polarisation.

**The Einstein-Cartan field equation.** For the totally antisymmetric torsion  $T^a_{bc} = (\tau/L)\varepsilon^a_{bc}$ , varying the Einstein-Cartan action with respect to the contorsion gives

$$\frac{\tau}{L} = \kappa^2 j_5, \quad \kappa^2 = \frac{8\pi}{M_P^2}, \quad (\text{C.6})$$

where the chiral current is the intensive chiral drive screened by the hypercharge polarisation,

$$j_5 = \frac{\langle \chi \rangle}{\Pi_{\text{ren}}} = \frac{N_L - N_R}{N_f \sum Y^2}. \quad (\text{C.7})$$

The geometric factor  $\kappa^2 L$  is evaluated exactly through the window-edge identity. The three-dimensional gravitational coupling  $\kappa^2$  is reduced from the four-dimensional Planck mass by the Kaluza-Klein volume factor, and combined with the radius  $L$  the factor reads

$$\kappa^2 L = \frac{M_G^2}{2\pi^2 M_P^2 k L^2} = \frac{1}{2\pi k L^4}, \quad (\text{C.8})$$

where the last step uses the window edge  $k L M_G = M_P \sqrt{\pi}$ , so  $M_G^2/(M_P^2 k L^2) = \pi/k L^4$ . The bare field equation carries the window geometry  $1/(2\pi k L^4)$ , which gives the bare torsion

$$\tau_{\text{bare}} = \frac{N_L - N_R}{2\pi k L^4}, \quad (\text{C.9})$$

and the hypercharge polarisation screens with the window capacity,

$$\text{screening} = \frac{2\pi k L^4}{N_f \sum Y^2}. \quad (\text{C.10})$$

The product closes on the content ratio

$$\tau = \tau_{\text{bare}} \cdot \text{screening} = \frac{N_L - N_R}{N_f \sum Y^2} = \frac{1}{50}. \quad (\text{C.11})$$

The window capacity cancels exactly between the bare field equation and the hypercharge screening, and the surviving value is the pure content ratio. Numerically the two factors are 0.004117 and 4.858, whose product is  $0.02 = 1/50$  exactly. The screening factor  $2\pi k L^4/(N_f \sum Y^2)$  is the non-perturbative spectral-sum vacuum polarisation: it is the discrete  $RP^3$  spectral sum evaluated in closed form (the window capacity  $2\pi k L^4$ ), not the standard one-loop continuous integral. The standard one-loop polarisation  $\Pi_{\text{1loop}} = (g_1^2/16\pi^2) \sum Y^2 \ln$  would need an unphysical  $\ln \approx 474$  to reach the content value; the framework instead evaluates the polarisation as the discrete spectral sum whose closed form is the window capacity, and this cancels exactly against the bare field equation to leave the content ratio  $\tau = 1/50$ . The polarisation is therefore the framework's own non-perturbative spectral-sum mechanism, not a perturbative loop. The torsion enters the spectrum only through  $\tau^2$ , so its effect is of order  $10^{-4}$ .

### C.3 Geometric dynamics

The torsion enters the physical content through the geometric quantities of the transverse-traceless sector. The  $J = 2$  Lichnerowicz eigenvalue of the Einstein-Cartan sector is

$$\lambda_{\text{EC}} = N_g \left(1 + \frac{\tau}{2}\right)^2 + 6 = 14 + 8\tau + 2\tau^2, \quad (\text{C.12})$$

the  $\mathfrak{su}(2)_L$  spin connection  $8(1 + \tau/2)^2$  dressed by the torsion plus the Lichnerowicz shift 6, and the transverse-traceless eigenvalue is

$$\lambda_{\text{TT}} = 14 = 2N_R, \quad (\text{C.13})$$

the content double of the seven right-handed singlets. The squash amplitude is the broken-generator content,

$$s_0 = n_{\text{broken}} \tau = 2\tau, \quad (\text{C.14})$$

with  $n_{\text{broken}} = \dim \text{SU}(2)_R - \dim \text{U}(1)_R = 3 - 1 = 2$  the two broken generators absorbed as the Goldstone directions of the  $W_R$  bosons, each contributing the torsion modulus. Its content ratio is

$$\frac{s_0}{N_R} = \frac{n_{\text{broken}}}{N_f \sum Y^2 N_R} = \frac{2}{15 \cdot (10/3) \cdot 7} = \frac{1}{175}, \quad (\text{C.15})$$

half the  $J = 2$  first-order torsion shift  $N_g \tau / 14 = 8\tau / 14$ . The  $2\pi$  Euclidean period appears throughout the framework as the common thread of the ultraviolet and infrared closures, in the amplitude  $e^{1/(2\pi)}$ , the window width  $2L = \sqrt{2\pi}$ , and the tensor ratio  $r = (1/2\pi)^2$ .

#### C.4 The gauge-coupling conservation law

The weak coupling at the emergence scale is the Killing normalisation of the isometry zero modes. The raw coupling is  $g_2^2 = 16\pi^2 / I_{kv}$  with the Killing integral  $I_{kv} = 2\pi^2 L^3$ , and the four-dimensional reduction inserts the normalisation factor  $(M_G / M_P)^2$ . At the emergence scale  $L = kL$  this gives

$$g_2^2 = \frac{16\pi^2 (M_G / M_P)^2}{2\pi^2 k L^3} = 8 \left( \frac{M_G}{M_P} \right)^2 k L^{-3}. \quad (\text{C.16})$$

The window-edge identity  $kL M_G = M_P \sqrt{\pi}$  collapses this to

$$g_2^2 = \frac{8\pi}{k L^5}, \quad \alpha_W = \frac{g_2^2}{4\pi} = \frac{2}{k L^5}. \quad (\text{C.17})$$

The bare Killing value sits +0.345% above the Standard Model value. The correction is the geometric-dynamics conservation law

$$\frac{1}{\alpha_{\text{SM}}} = \frac{1}{\alpha_W} + \frac{1}{N_c} - \frac{\tau^2 \pi}{2}, \quad (\text{C.18})$$

equivalently

$$N_c \left( \frac{1}{\alpha_{\text{SM}}} - \frac{1}{\alpha_W} + \frac{\tau^2 \pi}{2} \right) = 1. \quad (\text{C.19})$$

Here  $\alpha_{\text{SM}}$  is the physical weak coupling that the law predicts, the screened value computed from  $\alpha_W$  and the two correction terms, not an observed input. The law reads the window-capacity coupling  $\alpha_W$  and returns the physical coupling  $g_2(M_G) = \sqrt{4\pi \alpha_{\text{SM}}}$ . The two correction terms have independent physical origins. The colour-number term is the fermion screening of the gauge coupling, the ratio of the adjoint Casimir to the total fermion Dynkin index,

$$\frac{1}{N_c} = \frac{C_2(\text{SU}(2))}{\sum_f T(R_f)} = \frac{2}{6} = \frac{1}{3}, \quad (\text{C.20})$$

with  $C_2(\text{SU}(2)) = 2$  and  $\sum_f T(R_f) = 6$  the Dynkin sum of the three generations of doublets, so the screening is the content ratio one over the colour number. The torsion term  $-\tau^2 \pi / 2$  is the Einstein-Cartan correction of the geometric dynamics, the square of

the torsion modulus  $\tau^2 = 1/2500$  times the half-period  $\pi/2$ , the Euclidean causal-horizon factor  $T_{\text{eff}} = k/(2\pi)$  of the window. The torsion enters the couplings only through its square, because the totally antisymmetric torsion contributes to the kinetic term of the geometric dynamics at order  $\tau^2$ , so the correction is the torsion content ratio times the period factor. The bare coupling  $\alpha_W$  is the unscreened Killing value, and the physical coupling  $\alpha_{\text{SM}}$  is the screened value, which is why the physical coupling, not the bare one, stands on the left of the law. The colour-number term  $1/N_c$  and the torsion term  $\tau^2\pi/2$  remove the overshoot of the bare Killing value, and the resulting coupling is

$$g_2 = \sqrt{\frac{4\pi}{1/\alpha_W + 1/N_c - \tau^2\pi/2}} = 0.508848. \quad (\text{C.21})$$

The two correction terms are both content ratios of the framework, the colour number and the torsion squared, so the law has no free coefficient. The algebraic core of the law is the colour self-inverse  $N_c \cdot (1/N_c) = 1$  together with the cancellation of the torsion term, and the colour-number term is exactly the Casimir-over-Dynkin ratio  $C_2(\text{SU}(2))/\sum_f T(R_f)$  established above. The screening is the non-perturbative spectral-sum polarisation of Section C.2: the discrete  $RP^3$  spectral sum in closed form (the window capacity), not a standard one-loop vacuum-polarisation shift. The torsion factor carries a spectral-sum origin. Its explicit form is the product of three spectral factors,

$$\frac{\tau^2\pi}{2} = (2\tau^2)(2\pi)\xi = \frac{4\pi\tau^2}{N_g}, \quad (\text{C.22})$$

each traced to the spectrum. The factor  $2\tau^2$  is the second-order torsion term of the Einstein-Cartan curvature  $\lambda_{\text{EC}} = N_g(1 + \tau/2)^2 + 6 = 14 + 8\tau + 2\tau^2$  of Section C.2: the totally antisymmetric torsion  $T^a_{bc} = (\tau/L)\varepsilon^a_{bc}$  raises the scalar curvature by the six-component square  $2\tau^2$ . The factor  $2\pi$  is the Euclidean period, the Matsubara zero-frequency of the spectral sum, the same  $2\pi$  thread as  $\varepsilon = e^{1/2\pi}$ , the window capacity  $2\pi kL^4$  and  $g_A = 4/\pi$ . The factor  $\xi = (d-2)/(4(d-1)) = 1/8 = 1/N_g$  is the conformal coupling, the scalar-field curvature coupling of the frame whose duality  $N_g\xi = 1$  closes the law. The torsion mode's Euclidean spectral sum,  $2\tau^2 \cdot 2\pi$ , normalised by the generator count through  $\xi = 1/N_g$ , is the physical origin of the term, not an assigned constant. The two terms of the law therefore carry the same conformal coupling: the colour term  $1/N_c = N_g\xi/N_c$  and the torsion term  $\tau^2\pi/2 = 4\pi\tau^2\xi$  are both proportional to  $\xi$ , so the law is the conformal-gauge duality  $N_g\xi = 1$  written in the coupling-inverse basis, not an independent assumption. The hypercharge coupling follows from the same geometry through the squash mixing  $g_1 = g_2\kappa(2\tau)$  with  $\kappa^2(s) = (1+s)/(1-2s)^{5/2}$  at  $s = 2\tau$ , applied at the isometry-breaking scale. The mixing is the metric normalisation of the squashed internal space, not a fitted form. The isometry breaking  $\text{SU}(2)_R \rightarrow \text{U}(1)_R$  squashes the internal  $S^3$  along the  $\sigma_3$  axis, the  $\text{U}(1)_Y$  direction, giving the metric

$$ds^2 = (1+s)\sigma_1^2 + (1+s)\sigma_2^2 + (1-2s)\sigma_3^2, \quad (\text{C.23})$$

with eigenvalues  $\lambda_1 = \lambda_2 = 1+s$  and  $\lambda_3 = 1-2s$ . The volume ratio is  $V_3(s)/V_3(0) = \sqrt{\det g} = (1+s)\sqrt{1-2s}$ , and the  $\sigma_3$ -axis inverse metric is  $g^{33} = 1/(1-2s)$ . The  $\text{U}(1)_Y$

normalisation is the volume ratio times the third moment of the axis inverse metric,

$$\kappa^2(s) = \frac{V_3(s)}{V_3(0)} (g^{33})^3 = (1+s)\sqrt{1-2s}(1-2s)^{-3} = \frac{1+s}{(1-2s)^{5/2}}. \quad (\text{C.24})$$

The third moment is the Killing normalisation of the  $U(1)_Y$  gauge field on the squashed internal space: the  $F^2$  contraction of the  $\sigma_3$ -axis zero mode carries three inverse-metric factors, two from the field strength and one from the polarisation, so the normalisation integral is  $\int \sqrt{g} (g^{33})^3 d^3x$  normalised to its round value. At  $s = 0$  (round  $S^3$ )  $\kappa^2 = 1$ , so  $g_1 = g_2$ , the unification; at  $s = 2\tau$  the mixing is  $\kappa(2\tau) = 1.13183$ . The mixing coefficient is corrected by the content ratio

$$\delta_{g_1} = -\tau r_{23} \sum Y^2 \Delta_f \xi, \quad (\text{C.25})$$

with  $r_{23} = 3/(10\sqrt{3})$  the hypercharge-trace hierarchy ratio and the product  $\sum Y^2 \Delta_f \xi = (10/3)(3/2)(1/8) = 5/8$  the hypercharge capacity times the fermion conformal weight times the conformal coupling, the same conformal-gauge content that closes the  $1/N_c$  term above. The colour coupling carries the long-root bifurcation  $g_3 = g_2(1 + \alpha_{\text{GUT}}^2/K)$  with  $K = J(J+2)/d = 8/3$ , where  $J = 2$  is the squash angular momentum and  $d = 3$  the internal dimension, the kinetic eigenvalue  $J(J+2) = 8$  of the  $J = 2$  squash over the internal dimension, the long-root condensation diluting the two-loop self-coupling of the colour factor.

### C.5 Two-loop gauge coefficients

The renormalisation group running requires the gauge beta functions, and their coefficients are derived from the group-theoretic content rather than imported from a table. The five fermion representations are the framework's minimal combination, not a standard-model input. The chiral clause of the disorder axiom is carried by the two semispinor representations of  $\text{Spin}(4)$ , the left-handed  $(1/2, 0)$  an  $SU(2)_L$  doublet; the colour algebra is selected as the minimal non-trivial one ( $\mathfrak{su}(3)$ , with  $8 < 10 < 14$ ), and the fermion sits in its smallest irreducible representation, the fundamental  $3$ ; colour vectoriality (the  $SU(3)^3$  anomaly cancellation) puts the left-handed fermion in  $3$  and the right-handed in  $\bar{3}$ . The five representations are then chirality times colour assignment: the left doublet carries colour ( $Q_L$ ) or not ( $L_L$ ), and the right singlets mirror the doublet components,  $u_L, d_L \rightarrow u_R, d_R$  (colour  $\bar{3}$ ) and  $e_L \rightarrow e_R$  (colour  $1$ ), with  $\nu_L$  left Majorana. Hence two left doublets and three right singlets, the content  $N_L = 6 + 2 = 8 = N_g$  of Section 4.1. The uniqueness of these five follows from the same minimal combination. The chiral clause fixes the left sector as the single semispinor  $(1/2, 0)$ , an  $SU(2)_L$  doublet, and the colour assignment has exactly two options for this doublet: the minimal non-trivial  $3$  (giving  $Q_L$ ) or the trivial  $1$  (giving  $L_L$ ). The right sector is the Yukawa mirror of the left: each charged left component acquires exactly one right singlet,  $u_L, d_L \rightarrow u_R, d_R$  and  $e_L \rightarrow e_R$ , while the neutral  $\nu_L$  acquires none (Majorana). The counting is therefore two left doublets and three right singlets, exactly five, and no other representation is produced by the minimal combination. The scalar content is the colour-singlet zero mode of the framework, whose electroweak part is the doublet  $(1, 2)_{1/2}$ ; its hypercharge is fixed by the same charge convention, not imported.

The hypercharge assignment is derived next. Given the fermion representations and the Higgs hypercharge  $Y_H = 1/2$ , the charge convention  $Q = T_3 + Y$  gives the Yukawa relations

$$Y_u = Y_Q + \frac{1}{2}, \quad Y_d = Y_Q - \frac{1}{2}, \quad Y_e = Y_L - \frac{1}{2}. \quad (\text{C.26})$$

The mixed  $\text{SU}(2)^2\text{U}(1)$  anomaly vanishes when  $3Y_Q + Y_L = 0$ , and the gravitational anomaly when  $6Y_Q + 3Y_u + 3Y_d + 2Y_L + Y_e = 0$ . Substituting the Yukawa relations into these two equations gives  $3Y_Q - 1/2 = 0$ , hence  $Y_Q = 1/6$  and  $Y = (1/6, 2/3, -1/3, -1/2, -1)$ . The per-generation hypercharge moment is

$$\sum Y^2 = 6 \cdot \frac{1}{6^2} + 3 \cdot \frac{4}{9} + 3 \cdot \frac{1}{9} + 2 \cdot \frac{1}{4} + 1 \cdot 1 = \frac{10}{3}. \quad (\text{C.27})$$

The one-loop gauge coefficients follow from the group-theoretic content,

$$b_1 = \frac{2}{5}(3 \sum Y^2 + Y_H^2) = \frac{41}{10}, \quad (\text{C.28})$$

$$b_2 = -\frac{11}{3}C_2(\text{SU}(2)) + \frac{2}{3}T_f(\text{SU}(2)) + \frac{1}{3}T_H = -\frac{19}{6}, \quad (\text{C.29})$$

$$b_3 = -\frac{11}{3}C_2(\text{SU}(3)) + \frac{2}{3}T_f(\text{SU}(3)) = -7, \quad (\text{C.30})$$

with  $C_2(\text{SU}(2)) = 2$  and  $C_2(\text{SU}(3)) = N_c = 3$  the adjoint Casimirs,  $T_f(\text{SU}(2)) = T_f(\text{SU}(3)) = 6$  the three-generation Dynkin sums, and  $T_H = 1/2$  the Higgs doublet index. The two-loop matrix is derived from the same content. The diagonal entries are

$$B_{ii} = -\frac{34}{3}C_2(G_i)^2 + \left(2C_2^i(F) + \frac{10}{3}C_2(G_i)\right)S_2^i(F) + \left(2C_2^i(S) + \frac{1}{3}C_2(G_i)\right)S_2^i(S), \quad (\text{C.31})$$

and the off-diagonal entries are

$$B_{ij} = 2C_2^j(F)S_2^i(F) + 2C_2^j(S)S_2^i(S), \quad (\text{C.32})$$

with  $C_2$  the Casimir and  $S_2$  the Dynkin index of each representation, summed over the five Weyl states and the Higgs scalar. The Casimir and Dynkin index are defined as follows. The adjoint Casimir of  $\text{SU}(N)$  is  $C_2(\text{SU}(N)) = N$ , and the Dynkin index of a representation  $R$  is fixed by  $\text{Tr}(T_R^a T_R^b) = S_2(R)\delta^{ab}$ . The fundamental doublet of  $\text{SU}(2)$  has  $C_2 = 3/4$  and  $S_2 = 1/2$ , and the fundamental triplet of  $\text{SU}(3)$  has  $C_2 = 4/3$  and  $S_2 = 1/2$ . For the  $\text{U}(1)$  factor, in the GUT normalisation  $g_1 = \sqrt{5/3}g_Y$ , the Casimir and Dynkin index are  $C_2 = (3/5)Y^2$  and  $S_2 = (3/5)\sum Y^2$ .

The per-generation Weyl content, with the six entries  $(C_2^{\text{U}(1)}, C_2^{\text{SU}(2)}, C_2^{\text{SU}(3)}, S_2^{\text{U}(1)}, S_2^{\text{SU}(2)}, S_2^{\text{SU}(3)})$ , is

$$\begin{aligned} Q_L(3, 2)_{1/6} &: \left(\frac{3}{5}\frac{1}{36}, \frac{3}{4}, \frac{4}{3}, \frac{3}{5}\frac{1}{6}, \frac{3}{2}, 1\right), \\ u_R(3, 1)_{2/3} &: \left(\frac{3}{5}\frac{4}{9}, 0, \frac{4}{3}, \frac{3}{5}\frac{4}{3}, 0, \frac{1}{2}\right), \\ d_R(3, 1)_{-1/3} &: \left(\frac{3}{5}\frac{1}{9}, 0, \frac{4}{3}, \frac{3}{5}\frac{1}{3}, 0, \frac{1}{2}\right), \\ L_L(1, 2)_{-1/2} &: \left(\frac{3}{5}\frac{1}{4}, \frac{3}{4}, 0, \frac{3}{5}\frac{1}{2}, \frac{1}{2}, 0\right), \\ e_R(1, 1)_{-1} &: \left(\frac{3}{5}, 0, 0, \frac{3}{5}, 0, 0\right), \end{aligned} \quad (\text{C.33})$$

and the Higgs scalar  $(1, 2)_{1/2}$  contributes the real-scalar content  $(\frac{3}{5}\frac{1}{4}, \frac{3}{4}, 0, \frac{3}{5}, 1, 0)$ . The diagonal entry  $B_{11}$  is evaluated term by term. With  $C_2(G_1) = 0$  for the abelian factor,

$$B_{11} = 2 \sum_f C_2^{U(1)}(f) S_2^{U(1)}(f) n_g + 2 C_2^{U(1)}(H) S_2^{U(1)}(H), \quad (\text{C.34})$$

and inserting the hypercharge content, with  $\sum_f C_2^{U(1)} S_2^{U(1)} = (3/5)^2 \sum_f Y^4 = (3/5)^2 (\frac{1}{216} + \frac{16}{27} + \frac{1}{27} + \frac{1}{8} + 1)$  and  $n_g = 3$ , gives  $B_{11} = 199/50$ . The remaining entries follow by the same insertion. The full matrix is

$$B = \begin{pmatrix} 199/50 & 27/10 & 44/5 \\ 9/10 & 35/6 & 12 \\ 11/10 & 9/2 & -26 \end{pmatrix}. \quad (\text{C.35})$$

No entry of this matrix is taken from a table. The coefficients are the quadratic content ratios of the framework's fermion and scalar representations, each carried to its Casimir and Dynkin index.

The Yukawa-gauge mixing coefficients  $A_i$  are derived from the same content. The gauge beta carries the top-Yukawa term through the invariant  $Y_4(F) = (1/d(G_i)) \text{Tr}[C_2^{(i)}(F) Y^a Y^{+a}]$  with  $A_i = 2\kappa Y_4/y_t^2$  ( $\kappa = 1/2$  for Weyl fermions). The trace is the explicit matrix element

$$\text{Tr}[C_2^{(i)}(F) Y^a Y^{+a}] = 6 y_t^2 [C_2^{(i)}(Q_L) + C_2^{(i)}(u_R)], \quad (\text{C.36})$$

where the factor  $6 = 2 \times 3$  is the weak doublet times the colour triplet of  $Q_L$ . Each fermion block contributes the same 6: the Yukawa contraction  $Q_L^b H^a u_R \varepsilon_{ab}$  closes the SU(2) indices through  $\sum_a \varepsilon_{ab} \varepsilon_{ab'} = \delta_{bb'}$ , so the  $u_R$  trace carries the weak-contraction factor  $2 = \dim \text{SU}(2)$ . Hence

$$A_i = \frac{6}{d(G_i)} [C_2^{(i)}(Q_L) + C_2^{(i)}(u_R)], \quad (\text{C.37})$$

with  $d(G) = (1, 3, 8)$  and the Casimirs  $C_2^{(1)}(Q_L) = \frac{3}{5}\frac{1}{36}$ ,  $C_2^{(1)}(u_R) = \frac{3}{5}\frac{4}{9}$ ,  $C_2^{(2)}(Q_L) = \frac{3}{4}$ ,  $C_2^{(2)}(u_R) = 0$ ,  $C_2^{(3)}(Q_L) = C_2^{(3)}(u_R) = \frac{4}{3}$ . The three entries are

$$A = \left( \frac{17}{10}, \frac{3}{2}, 2 \right), \quad (\text{C.38})$$

reproducing the standard two-loop values from the SM content alone, with no external table.

## C.6 The top-Yukawa and Higgs-quartic beta functions

The scalar-sector running uses the top-Yukawa and Higgs-quartic beta functions, evaluated on the same content by the general two-loop formulas of [17] specialised to the standard-model content in [18]. The conventions here are  $V = -m^2|H|^2 + \lambda|H|^4$  and the GUT normalisation  $g_1 = \sqrt{5/3} g_Y$ ; [18] writes the potential as  $m^2 \varphi^\dagger \varphi + (\lambda_{LX}/2)(\varphi^\dagger \varphi)^2$ , so the

rescaling  $\lambda = \lambda_{LX}/2$ ,  $\beta_\lambda = \beta_{\lambda,LX}/2$  (a pure coupling rescaling) converts their results to the convention used here. The content enters through the trace invariants

$$Y_2(S) = \text{Tr}[3H^\dagger H] = N_c y_t^2, \quad H(S) = \text{Tr}[3(H^\dagger H)^2] = N_c y_t^4, \quad \chi_4(S) = \frac{9}{4} \text{Tr}[3(H^\dagger H)^2] = \frac{9}{4} N_c y_t^4, \quad (\text{C.39})$$

and

$$Y_4(S) = \left( \frac{17}{20} g_1^2 + \frac{9}{4} g_2^2 + 8 g_3^2 \right) y_t^2, \quad (\text{C.40})$$

where the gauge coefficients are themselves the one-loop content of Section C.5, the anti-commutator  $-3g^2\{C_2(F), Y\}$  evaluated per gauge group,

$$\frac{17}{20} = 3[C_2^{\text{U}(1)}(Q_L) + C_2^{\text{U}(1)}(u_R)] = 3\left(\frac{3}{5} \frac{1}{36} + \frac{3}{5} \frac{4}{9}\right), \quad \frac{9}{4} = 3 C_2^{\text{SU}(2)}(Q_L) = 3 \cdot \frac{3}{4}, \quad 8 = 3[C_2^{\text{SU}(3)}(Q_L) + C_2^{\text{SU}(3)}(u_R)] \quad (\text{C.41})$$

The family count  $n_g = 3$  entering below is the window-capacity generation number of Section 4.1.

The one-loop top Yukawa is

$$\frac{\beta_{y_t}^{(1)}}{y_t} = \left( \frac{3}{2} + N_c \right) y_t^2 - \frac{17}{20} g_1^2 - \frac{9}{4} g_2^2 - 8 g_3^2 = \frac{9}{2} y_t^2 - \frac{17}{20} g_1^2 - \frac{9}{4} g_2^2 - 8 g_3^2, \quad (\text{C.42})$$

the Yukawa self-coupling  $3/2$  plus the colour trace  $Y_2(S) = N_c y_t^2$ ; the U(1) coefficient  $17/20$  is in the GUT normalisation, i.e.  $-(17/12)g_Y^2$  in the hypercharge coupling. The two-loop top Yukawa is

$$\begin{aligned} \frac{\beta_{y_t}^{(2)}}{y_t} = & -12 y_t^4 + \left( 36 g_3^2 + \frac{225}{16} g_2^2 + \frac{393}{80} g_1^2 \right) y_t^2 - 108 g_3^4 - \frac{23}{4} g_2^4 + \frac{1187}{600} g_1^4 \\ & + 9 g_2^2 g_3^2 + \frac{19}{15} g_1^2 g_3^2 - \frac{9}{20} g_1^2 g_2^2 + 6 \lambda^2 - 12 \lambda y_t^2, \end{aligned} \quad (\text{C.43})$$

with the content structure

$$\begin{aligned} -12 = \frac{3}{2} - \frac{9}{4} N_c - \frac{9}{4} N_c, \quad 36 = \frac{5}{2} \cdot 8 + 16, \quad \frac{225}{16} = \frac{5}{2} \cdot \frac{9}{4} + \frac{135}{16}, \quad \frac{393}{80} = \frac{5}{2} \cdot \frac{17}{20} + \frac{223}{80}, \\ \frac{1187}{600} = \frac{9}{200} + \frac{29}{45} n_g, \quad -\frac{23}{4} = -\left( \frac{35}{4} - n_g \right), \quad -108 = -\left( \frac{404}{3} - \frac{80}{9} n_g \right), \end{aligned} \quad (\text{C.44})$$

where the first line is the  $\chi_4$ - and  $Y_4$ -content of the trace invariants above, and the remaining coefficients (16, 135/16, 223/80, 9/200, 29/45, 35/4, 404/3, 80/9, 9, 19/15, 9/20, 6, 12) are the universal two-loop numbers of the general evaluation. The one-loop Higgs quartic is

$$\beta_\lambda^{(1)} = 24 \lambda^2 - 3 \lambda \left( 3 g_2^2 + \frac{3}{5} g_1^2 \right) + \frac{3}{8} \left( 2 g_2^4 + (g_2^2 + \frac{3}{5} g_1^2)^2 \right) - 6 y_t^4 + 12 \lambda y_t^2, \quad (\text{C.46})$$



whose Yukawa part is the content combination  $4Y_2(S)\lambda - 4H(S) = 12\lambda y_t^2 - 6y_t^4$ . The full two-loop Higgs quartic is

$$\begin{aligned}\beta_\lambda^{(2)} = & -312\lambda^3 + 36\lambda^2\left(3g_2^2 + \frac{3}{5}g_1^2\right) - 144\lambda^2 y_t^2 - \frac{73}{8}\lambda g_2^4 + \frac{117}{20}\lambda g_2^2 g_1^2 + \frac{1887}{200}\lambda g_1^4 \\ & + \left(\frac{17}{2}g_1^2 + \frac{45}{2}g_2^2 + 80g_3^2\right)\lambda y_t^2 - 3\lambda y_t^4 + 30y_t^6 \\ & - 32g_3^2 y_t^4 - \frac{8}{5}g_1^2 y_t^4 - \frac{9}{4}g_2^4 y_t^2 + \frac{63}{10}g_2^2 g_1^2 y_t^2 - \frac{171}{100}g_1^4 y_t^2 \\ & + \frac{305}{16}g_2^6 - \frac{289}{80}g_2^4 g_1^2 - \frac{1677}{400}g_2^2 g_1^4 - \frac{3411}{2000}g_1^6,\end{aligned}\tag{C.47}$$

with the content structure

$$-312 = -78 \cdot 4, \quad 108 = 54 \cdot 2, \quad -144 = -24 \cdot Y_2(S)/y_t^2 \cdot 4, \quad \frac{17}{2} = 10 \cdot \frac{17}{20}, \quad \frac{45}{2} = 10 \cdot \frac{9}{4}, \quad 80 = 10 \cdot 8,\tag{C.48}$$

$$30 = \frac{60}{2}, \quad -32 = -\frac{64}{2}, \quad -\frac{8}{5} = -\frac{16}{5}/2, \quad -\frac{9}{4} = -\frac{9}{2}/2, \quad \frac{63}{10} = \frac{63}{5}/2, \quad -\frac{171}{100} = -\frac{171}{50}/2,\tag{C.49}$$

$$\frac{305}{16} = \left(\frac{497}{8} - 8n_g\right)/2, \quad -\frac{289}{80} = -\left(\frac{97}{40} + \frac{8}{5}n_g\right)/2, \quad -\frac{1677}{400} = -\left(\frac{717}{200} + \frac{8}{5}n_g\right)/2, \quad -\frac{3411}{2000} = -\left(\frac{531}{1000} + \frac{24}{25}n_g\right),\tag{C.50}$$

$$-\frac{73}{8} = -\left(\frac{313}{8} - 10n_g\right), \quad \frac{1887}{200} = \frac{687}{200} + 2n_g,\tag{C.51}$$

where the  $\lambda y_t^2 g^2$  terms are the  $10\lambda Y_4(S)$  content, the  $y_t^6$ ,  $g^2 y_t^4$ ,  $g^4 y_t^2$  terms are the fermionic traces evaluated on the top content, and the  $n_g$ -carrying coefficients are the fermionic Dynkin sums over the three generations. Every coefficient above is displayed as a content combination (or a documented universal two-loop number); none is imported from a table, and each rational identity is elementary (the displayed equalities are the decompositions).

### C.7 The Landau-Zener ladder

The generation masses are suppressed exponentially by the extrusion of the modes  $n = \{0, 2, 4\}$  under the non-adiabatic squeezing of the scale flow. The extrusion is a Landau-Zener two-level crossing in the scale flow  $L(k) = C/k$ , with the avoided-crossing Hamiltonian

$$H(t) = \begin{pmatrix} vt & \Delta \\ \Delta & -vt \end{pmatrix}, \quad P = e^{-2\pi\Delta^2/v},\tag{C.52}$$

and the LZ index  $\gamma = 2\pi\Delta^2/v$  decomposes into the adiabatic window width and the non-adiabatic torsion. The mass ratios are  $m_i \propto e^{-\alpha n_i}$ , and the sector index is fixed by the internal ladder. The up-sector index is the adiabatic window width minus the non-adiabatic torsion correction,

$$\alpha_{\text{up}} = kL - 2\tau = 2.4535,\tag{C.53}$$

where  $kL$  is the adiabatic part (the extrusion of a mode across the full coarse-graining window) and  $2\tau = n_{\text{broken}}\tau$  is the non-adiabatic deficit (the Einstein-Cartan torsion of the

two broken generators, each carrying  $\tau$ ), so the top-to-charm ratio is  $m_t/m_c = e^{2\alpha_{\text{up}}}$ , and the difference  $kL - \alpha_{\text{up}} = 2\tau$  is exact. The down and lepton sectors descend by the step

$$\Delta = 6(1 - n_s)kL_{\text{CMB}}, \quad (\text{C.54})$$

the product of the six generators of the  $\mathfrak{so}(4)$  isometry, the spectral tilt  $1 - n_s$ , and the microwave-pivot window  $kL_{\text{CMB}}$ . The down and lepton steps are split by the  $9/8$  hypercharge identity

$$\frac{9}{8} = \frac{1}{1 - (Y_d/Y_l)^2}, \quad (\text{C.55})$$

with  $Y_d = 1/3$  and  $Y_l = 1$ , so  $(Y_l^2 - Y_d^2)/Y_l^2 = 8/9$ , the lepton step carrying  $8/9$  of the down step. Writing the lepton step as  $s$  and the down step as  $(9/8)s$ , and using that the lepton rung spans two steps,  $(9/8)s + s = 2\Delta$ , one obtains  $s = 16\Delta/17$  and the down step  $(9/8)(16\Delta/17) = 18\Delta/17$ . The indices are

$$\alpha_{\text{dn}} = \alpha_{\text{up}} - \frac{18}{17}\Delta, \quad \alpha_{\text{lp}} = \alpha_{\text{up}} - 2\Delta. \quad (\text{C.56})$$

The first-generation step of the down sector carries the colour dilution

$$\alpha_{\text{sd}} = \alpha_{\text{dn}} - \frac{kL_{\text{CMB}}}{6}, \quad (\text{C.57})$$

the microwave window divided by the six generators of  $\mathfrak{so}(4)$ , the colour dilution of the first-generation extrusion. The symmetry form of this step is  $\alpha_{\text{sd}} = \Delta_f(1 - s_0/N_R) = (3/2)(1 - 2\tau/7)$ , the fermion conformal weight  $\Delta_f = d/2 = 3/2$  times the gravity correction  $1 - s_0/N_R$  with  $s_0/N_R = 1/175$ .

The four coefficients share one physical origin, the non-adiabatic extrusion of the generation modes. The up index is the window width with the torsion subtracted,  $kL - \alpha_{\text{up}} = 2\tau$  exactly, because the Einstein-Cartan torsion is the non-adiabatic part of the scale flow, and the LZ exponent is the adiabatic window width minus the non-adiabatic torsion; the factor 2 is the number of broken generators of the squash,  $n_{\text{broken}} = 2$ . The step  $\Delta$  is the extrusion rate of the sector ladder, the product of the six  $\mathfrak{so}(4)$  generators (the extrusion coupling, the isometry of the internal space), the spectral tilt  $1 - n_s$  (the closed window-evolution rate, itself  $\tau \cdot 7/4$ ), and the microwave window  $kL_{\text{CMB}}$  (the scale at which the extrusion is observed). The  $9/8$  split is the hypercharge weight in the covariant derivative on  $RP^3$  shifting the LZ exponent: the down and lepton doublets carry different hypercharge weights  $Y_d = 1/3$  and  $Y_l = 1$ , so the lepton rung carries  $8/9$  of the down rung, the exact identity  $9/8 = 1/(1 - (Y_d/Y_l)^2)$ . The colour dilution  $kL_{\text{CMB}}/6$  is the first-generation extrusion divided by the six generators, the content of the first-generation mode spread over the full  $\mathfrak{so}(4)$  isometry. None of the four coefficients is a fitted rule; each is a content ratio of the non-adiabatic extrusion.

### C.8 The top-quark anchor

The top Yukawa is the  $(0, 0)$  diagonal overlap of the  $n = 0$  spinor mode with the scalar condensate. The overlap is normalised to unity,

$$y_0 = 1, \quad (\text{C.58})$$

because the mass operator is the  $(0,0)$  scalar of  $\text{SO}(4)$ , the torsion singlet  $T_{abc}\gamma^{abc}$ , and its matrix element is evaluated explicitly on the generation modes. A generation mode  $n$  carries the  $\text{SO}(4)$  content  $(j_L, j_R) = ((n+1)/2, n/2)$ , so the top  $n=0$  is  $(j_L, j_R) = (1/2, 0)$ . The matrix element of the  $(0,0)$  scalar between a mode and itself is the product of the two  $\text{SU}(2)$  scalar-channel Clebsch-Gordan coefficients,

$$y_0 = \langle (j_L, j_R) | (0,0) | (j_L, j_R) \rangle = C(j_L, 0, j_L) C(j_R, 0, j_R), \quad (\text{C.59})$$

where  $C(j, 0, j) = 1$  is the  $\text{SU}(2)$  Clebsch-Gordan coefficient of the scalar channel: the identity representation couples  $j \otimes 0 = j$  with the unique coefficient one. Hence

$$y_0 = C(\tfrac{1}{2}, 0, \tfrac{1}{2}) C(0, 0, 0) = 1 \cdot 1 = 1. \quad (\text{C.60})$$

This is the explicit wave-function matrix element, evaluated on the  $n=0$  generation mode. The same element is 1 for every generation mode,  $C(j_L, 0, j_L) C(j_R, 0, j_R) = 1$  (the scalar channel couples each mode to itself with unit coefficient), so the generations differ only through the ladder of Section C.7, not through this element. It is also scale-invariant, in the sense that the geometric renormalisation group leaves it unchanged, so  $y_0$  does not run. The top mass is then fixed by the vacuum expectation value of Section 7.3, which is itself derived from the emergence scale through the window-squared line, so  $m_t = y_0 v / \sqrt{2}$  carries no observed mass and no free Yukawa coupling. The lower generations descend by the ladder of Section C.7. The bottom absolute mass is the geometric mean of the strange and the top dressed by the window evolution,

$$\frac{y_b}{y_t} = e^{-(2\alpha_{\text{dn}} - n_s^{\text{tilt}}(kL_{\text{CMB}} + 2\tau))}, \quad (\text{C.61})$$

$$\text{so } m_b^2 = m_s m_t e^{n_s^{\text{tilt}}(kL_{\text{CMB}} + 2\tau)}.$$

### C.9 Charged-fermion masses

The exponent 20 can be understood systematically as a heat-kernel coefficient. For the Laplacian on  $RP^3$ , the Seeley–deWitt expansion gives  $\text{Tr } e^{-t\Delta_{RP^3}} \sim a_0 t^{-3/2} + a_1 t^{-1} + \dots$ , where the first non-trivial coefficient is  $a_1 = (d+1)(\sum Y^2 \Delta_f) = 4 \times 5 = 20$ . This is the same coefficient structure that appears in the spectral action principle of Chamseddine and Connes [4], specialised to the present spectral datum. The closed form  $m_e = M_P e^{-20kL}(1 - s_0\kappa)$  is the evaluation of this heat-kernel coefficient on the scale trajectory  $L(k) = C/k$ , Wick-rotated to the physical mass. The exponent is therefore not an ad hoc product but the first non-trivial heat-kernel invariant of the spectral datum.

The electron mass is the Planck-anchored exponential chain with the low-scale squash correction,

$$m_e = M_P e^{-20kL}(1 - s_0\kappa), \quad (\text{C.62})$$

where the exponent  $20 = 4 \times 5$  is the structural counting of the fourfold and fivefold content, and  $s_0\kappa$  is the squash correction of Section C.4. The two factors have independent first-principles origins. The factor  $4 = d+1$  is the internal dimension  $d=3$

plus one, the four levels of the spectral cascade. The factor  $5 = \sum Y^2 \Delta_f$  is the hypercharge capacity  $\sum Y^2 = 10/3$  times the fermion conformal weight  $\Delta_f = d/2 = 3/2$ , which equals the five fermion representations of one generation,  $Q_L, u_R, d_R, L_L, e_R$ , the content of Section 4.1. The product, not the sum, appears because each of the four cascade levels acts on the complete content of one generation, the five species, so the exponent is  $20 = (d+1)(\sum Y^2 \Delta_f) = 4 \times 5$ , a pure content ratio. The cascade form of the electron mass is

$$m_e = y_0 O_e \frac{v_{\text{dil}}}{\sqrt{2}}, \quad O_e = 1 - \frac{1}{1 + kL^2} e^{-1/kL}, \quad (\text{C.63})$$

the universal Yukawa seed  $y_0 = 1$  times the  $(0,0)$  overlap  $O_e$  of the electron  $l = 0$  mode with the dilaton scalar, times the dilaton vacuum expectation value  $v_{\text{dil}}$  at the electron scale. The overlap is the explicit constant-mode integral,

$$O_e = \int_{RP^3} d^3y \sqrt{g} \bar{\chi}_0(y) \varphi_{\text{dil}}^{(0)}(y) \chi_0(y), \quad (\text{C.64})$$

where  $\chi_0 = 1/\sqrt{2\pi^2 a^3}$  is the  $l = 0$  constant spinor and  $\varphi_{\text{dil}}^{(0)} = v_{\text{dil}}/\sqrt{2\pi^2 a^3}$  the  $l = 0$  constant dilaton, both spatially uniform, so the volume integral cancels the normalisation factors, giving  $O_e = 2\pi^2 a^3 (1/\sqrt{2\pi^2 a^3})(v_{\text{dil}}/\sqrt{2\pi^2 a^3})(1/\sqrt{2\pi^2 a^3}) = v_{\text{dil}}/\sqrt{2\pi^2 a^3}$ , i.e.  $O_e = 1$  after normalising by  $v_{\text{dil}}/\sqrt{2\pi^2 a^3}$  (the two constant modes overlap exactly, the unit  $(0,0)$  Clebsch-Gordan weight). The small deficit  $\delta(kL) = \frac{1}{1+kL^2} e^{-1/kL}$  is the finite-size back-reaction: the dilaton profile has a weak  $kL$ -dependent gradient on  $RP^3$ , and  $\delta(kL)$  is its first correction, so  $O_e = 1 - \delta(kL)$ .

The exponential chain  $m_e = M_P e^{-20kL}$  is the compressed statement of this cascade, and the equivalence is made explicit as follows. The electron-scale dilaton VEV descends from the Planck scale through the spectral cascade,

$$v_{\text{dil}} = M_P e^{-20kL} \frac{\sqrt{2}(1 - s_0\kappa)}{y_0 O_e}, \quad (\text{C.65})$$

so that the cascade form reproduces the exponential chain exactly,

$$m_e = y_0 O_e \frac{v_{\text{dil}}}{\sqrt{2}} = M_P e^{-20kL} (1 - s_0\kappa). \quad (\text{C.66})$$

The descent exponent  $20 = (d+1)(\sum Y^2 \Delta_f) = 4 \cdot 5$  is the KK cascade: the dilaton (the scalar zero mode) descends through  $d+1 = 4$  levels (the internal dimension 3 plus the scale-flow direction), each level acting on the complete one-generation content  $\sum Y^2 \Delta_f = (10/3)(3/2) = 5$  species; each (level  $\times$  species) unit crosses the coarse-graining window once and is suppressed by the LZ survival factor  $e^{-kL}$  (one window width  $kL$ , the same window whose circumference  $4\pi kL$  sets the electroweak hierarchy  $\ln(M_G/v)$  of Section C.11). The four steps therefore give  $(e^{-kL})^{4 \cdot 5} = e^{-20kL}$ . The  $O(1)$  prefactor  $\sqrt{2}(1 - s_0\kappa)/(y_0 O_e)$  carries the normalisation (the Yukawa factor  $\sqrt{2}$  and the seed  $y_0 = 1$ ) and the two level-corrections — the finite-size back-reaction  $\delta$  inside  $O_e$  and the  $J = 2$  squash  $s_0\kappa$  — both  $O(1)$  corrections to the dominant exponential, in the same two-correction pattern as the electroweak hierarchy  $\ln(M_G/v) = 4\pi kL - \ln(3\alpha/\pi) + s_0\kappa$ . The compression is thus explicit:

the  $4 \times 5 = 20$  window crossings collapse into the single exponent  $20kL$ , and the  $O(1)$  factor is the residual normalisation plus the two corrections. The muon-to-electron ratio carries the Euclidean period,

$$\frac{m_\mu}{m_e} = e^{2\alpha_{1p} + \sqrt{2\pi}}, \quad (\text{C.67})$$

the lepton LZ index plus the entropy-minimum distance  $2L = \sqrt{2\pi}$ .

### C.10 Neutrino masses and the PMNS matrix

The heaviest neutrino mass is set by the Weinberg dimension-five operator with the two-period family scale,

$$m_{\nu_3} = \frac{v^2(2\pi)^2}{k_{\text{GUT}}}(1 + s_0\kappa) = 0.0502 \text{ eV}, \quad (\text{C.68})$$

the electroweak scale squared times the Euclidean-period factor divided by the breaking scale, with the squash correction  $s_0\kappa$  of Section C.4. The form is the dimension-five operator  $\mathcal{L} = c(LH)(LH)/\Lambda$  with the seesaw family scale  $\Lambda = k_{\text{GUT}}/(2\pi)^2$ , giving  $m_{\nu_3} = cv^2/\Lambda$  with  $c = (2\pi)^2$ . The  $(2\pi)^2$  is the Euclidean period squared, the same  $2\pi$  thread as the amplitude  $e^{1/(2\pi)}$  and the window width  $2L = \sqrt{2\pi}$ . The lighter masses descend by the hypercharge-trace hierarchy ratios,

$$\frac{m_{\nu_1}}{m_{\nu_2}} = \frac{1}{\text{Tr}(Y^2)} = \frac{3}{10}, \quad \frac{m_{\nu_2}}{m_{\nu_3}} = \frac{1}{\sqrt{3} \text{Tr}(Y^2)} = \frac{3}{10\sqrt{3}}, \quad (\text{C.69})$$

so  $m_{\nu_1} = m_{\nu_3}r_{12}r_{23}$  with  $r_{12} = 3/10$  and  $r_{23} = 3/(10\sqrt{3})$ . The ratios are not assigned by hand; they are the eigenvalue ratios of an explicitly constructed mass matrix. The effective matrix is the spectral assembly

$$M_\nu = U \text{diag}(\lambda_3, \lambda_2, \lambda_1) U^\dagger \frac{v^2}{\Lambda}, \quad \lambda = \left\{ 1, \frac{1}{\sqrt{3} \text{Tr}(Y^2)}, \frac{1}{\sqrt{3} \text{Tr}(Y^2)^2} \right\}, \quad (\text{C.70})$$

where  $U$  is the PMNS matrix of the framework's content-ratio angles (Section C.10 below) and the eigenvalues are the inverse powers of the first non-zero hypercharge moment. Diagonalising  $M_\nu$  returns the hierarchy as the eigenvalue ratios,  $m_1/m_2 = \lambda_1/\lambda_2 = 1/\text{Tr}(Y^2) = 3/10$  and  $m_2/m_3 = \lambda_2/\lambda_3 = 1/(\sqrt{3} \text{Tr}(Y^2))$ , to machine precision ( $10^{-13}$ ). The hierarchy is therefore the eigenvalue structure of the hypercharge-trace flavour matrix, the content ratio of the same trace that closes the gauge sector, and no mass is inserted by hand. The  $\sqrt{3}$  is the internal-space geometry factor  $\sin(\pi/3) \cdot 2$  that also enters the reactor angle.

The PMNS mixing angles carry the two-period imprint. The solar angle is the neutrino mass ratio,

$$\sin^2 \theta_{12} = \frac{m_{\nu_1}}{m_{\nu_2}} = \frac{3}{10} = 0.30, \quad (\text{C.71})$$

the light-pair mixing of the two-state texture. In the near-tribimaximal texture the two lightest states decouple from the heavy state, and the mixing of the light pair is the ratio of their masses,  $\sin^2 \theta_{12} = m_{\nu_1}/m_{\nu_2}$ , the standard two-state seesaw relation. The reactor angle is the Euclidean-period imprint times the internal-space geometry factor,

$$\sin^2 \theta_{13} = \left( \frac{1}{2\pi} \right)^2 \frac{\sqrt{3}}{2} = 0.02194, \quad (\text{C.72})$$

and the atmospheric angle is

$$\sin^2 \theta_{23} = \frac{1}{2} + \frac{\text{Tr}(T_3^2)}{(2\pi)^2} = 0.5507. \quad (\text{C.73})$$

All three angles are fixed by the same content ratios that fix the charged sector, so the PMNS matrix is closed within the framework. The solar angle is the neutrino mass ratio, the hypercharge-trace reciprocal of Section C.10. The atmospheric angle is the maximum-mixing value  $1/2$  corrected by the isospin trace over the Euclidean period,  $\text{Tr}(T_3^2) = 2$ . The reactor angle is the period imprint  $1/(2\pi)^2$  times the internal-space geometry factor  $\sqrt{3}/2 = \sin(\pi/3)$ , the  $S^3 \rightarrow RP^3$  quotient projection. The structure is the near-tribimaximal form dressed by the two-period imprint, and no angle carries a free parameter.

### C.11 The electroweak order parameter

The isometry breaking is driven by the  $J = 2$  squash mode, the order parameter  $\varphi$  of the internal geometry. Its dynamics is the Landau potential on the curvature axis,

$$V(\varphi) = \frac{1}{2}\xi(R - R_c)\varphi^2 + \frac{\lambda}{4}\varphi^4, \quad (\text{C.74})$$

with  $\xi = 1/8$  the conformal coupling and  $R_c = 6/\pi$  the critical curvature. The effective mass squared is  $m^2 = \xi(R - R_c)$ , tachyonic for  $R < R_c$ , so the condensation is triggered by the curvature crossing, not by a free-spectrum instability. The quartic coupling is fixed by the stationarity of the potential at the squash vacuum  $\varphi_0 = 2\tau$ ,

$$\frac{\partial V}{\partial \varphi} = 0 \quad \Rightarrow \quad \lambda = \frac{\xi(R_c - R_{\text{GUT}})}{(2\tau)^2} = 149, \quad (\text{C.75})$$

with  $R_{\text{GUT}} = 6/L_{\text{GUT}}^2$  and  $L_{\text{GUT}} = \sqrt{3}/\tau$  the isometry-breaking scale. The Einstein-Cartan torsion algebra fixes the three coefficients of the parity-preserving torsion Lagrangian

$$L_{\text{torsion}} = aT^2 + bT^{bac}T_{abc} + c(T^a_{ab})^2. \quad (\text{C.76})$$

The algebraic-torsion condition, the requirement that the torsion field equation be algebraic rather than dynamical, fixes  $b = 4a$ , the Holst-Immirzi condition, and the trace term fixes  $c = -(a + b/3) = -(7/3)a$ . The remaining coefficient  $a = M_{\text{G}}^3/4$  is fixed by the canonical normalisation of the torsion kinetic term at the emergence scale, and the stationarity  $\partial V/\partial \varphi = 0$  at  $\varphi_0 = 2\tau$  then selects the quartic coupling. The vacuum expectation value is the emergence scale times the window-squared line,

$$v = M_{\text{G}} \frac{3\alpha}{\pi} e^{-4\pi kL} (1 - s_0\kappa), \quad (\text{C.77})$$

the exponential  $e^{-4\pi kL}$  carrying the hierarchy  $v/M_{\text{G}} \sim 10^{-16}$ , with the squash correction  $s_0\kappa$  of Section C.4. The two factors of the line have independent origins. The prefactor  $3\alpha/\pi$  is the product of three pieces, the  $J = 2$  squash bifurcation  $3\alpha/2$  with  $\alpha = 1/(16\pi^2)$  the one-loop weight, the Fourier prefactor  $1/\pi$  of the window, and the factor 2 from the double crossing,  $(3\alpha/2)(1/\pi)(2) = 3\alpha/\pi$ . The exponential  $e^{-4\pi kL}$  is the Landau-Zener survival of the  $J = 2$  mode crossing the coarse-graining window twice, once at creation and once at stabilisation, each crossing suppressing by  $e^{-2\pi kL}$ , the same exponential that governs the curvature amplitude of Section C.13.

### C.12 CKM mixing and CP

The quark-mixing elements follow from the mass ladder through the Gatto relation. The first-generation ratio carries the hypercharge first-generation factor  $|Y_d|/|Y_u| = 1/2$ ,

$$\frac{m_d}{m_s} = e^{-2\alpha_{\text{dn}}} \left(1 + \frac{|Y_d|}{|Y_u|}\right)^2 = e^{-2\alpha_{\text{dn}}} \left(\frac{3}{2}\right)^2, \quad (\text{C.78})$$

and the Cabibbo element is the Gatto dominant term,

$$|V_{us}| = \sqrt{\frac{m_d}{m_s}} = 0.2237. \quad (\text{C.79})$$

The cross-generation elements are

$$|V_{cb}| = \frac{1}{2}e^{-\alpha_{\text{up}}}(1 - s_0\kappa), \quad |V_{ub}| = \frac{1}{2}e^{-2\alpha_{\text{up}}}(1 + \tau\kappa), \quad (\text{C.80})$$

the adjacent-generation element carrying the squash correction  $s_0\kappa$  and the cross-generation element the chiral correction  $\tau\kappa$ . The three-element product closes on the weak cube,

$$|V_{us}||V_{cb}||V_{ub}| = \alpha_W^3, \quad (\text{C.81})$$

so the geometric mean of the three cross-generation mixings equals the weak coupling at the electroweak scale, a colour-diluted weak cube.

The CKM phase is the colour-number dilution of the left-right content ratio. The left-right ratio is

$$\frac{N_L}{N_R} = \frac{8}{7}, \quad (\text{C.82})$$

eight left-handed doublets against seven right-handed singlets per generation. The lepton PMNS phase carries the undiluted ratio  $\delta_{\text{PMNS}} \sim (8/7)\pi$ , and the quark phase is diluted by the colour number,

$$\delta_{\text{CKM}} = \frac{8}{7} \frac{\pi}{N_c} = \frac{8\pi}{21} = 68.57^\circ. \quad (\text{C.83})$$

The Jarlskog invariant is the exact formula with the closed elements and the derived phase,

$$J = |V_{us}||V_{cb}||V_{ub}| c_{12}c_{23} \sin \delta = 3.15 \times 10^{-5}. \quad (\text{C.84})$$

The baryon asymmetry is the Sakharov content. The Jarlskog invariant carries the weak cube through the three-element product, so with the two weak sphaleron vertices the total power is the standard sphaleron rate,

$$\eta_B = \frac{J\alpha_W^2}{56} = \frac{\alpha_W^5 c_{12}c_{23} \sin \delta}{56} = 6.09 \times 10^{-10}, \quad (\text{C.85})$$

the five weak factors being the three CKM mixings and the two sphaleron vertices, and  $1/56 = 1/(N_L N_R)$  the content count. The factor  $56 = N_L N_R = 8 \times 7$  is the product of the left and right content counts of one generation, and it decomposes as  $1/56 = \xi/n_R$ , the conformal-gauge duality  $\xi = 1/N_g = 1/8$  times the inverse right-handed count  $1/n_R = 1/7$ , the same content-ratio structure that closes the  $1/N_c$  term of the gauge conservation law of Section C.4. The Sakharov out-of-equilibrium condition is therefore the framework's content-count normalisation, not a fitted rate: the baryon asymmetry is the standard sphaleron power  $\alpha_W^5$  (three CKM mixings times two weak sphaleron vertices) divided by the content count  $N_L N_R$ .

### C.13 Cosmology

The dark-energy density is the lightest-neutrino mass floor,

$$\rho_\Lambda = Y_u m_{\nu_1}^4 = \frac{2}{3} m_{\nu_1}^4, \quad (\text{C.86})$$

where  $Y_u = 2/3$  is the up-quark hypercharge and the lightest neutrino mass is derived from the Weinberg operator

$$m_{\nu_3} = \frac{v^2(2\pi)^2}{k_{\text{GUT}}}, \quad m_{\nu_1} = m_{\nu_3} \cdot \frac{m_1}{m_2} \cdot \frac{m_2}{m_3}, \quad (\text{C.87})$$

with the hierarchy ratios  $m_1/m_2 = 1/\text{Tr}(Y^2) = 3/10$  and  $m_2/m_3 = 1/(\sqrt{3}\text{Tr}(Y^2))$ . The physical content of this relation is that the lightest fermion sets the vacuum-energy floor. The neutrino is the lightest fermion of the spectrum, and its mass is the smallest nonzero energy scale the spectrum carries, so the vacuum energy is anchored at that floor rather than at an arbitrary scale. The weight  $Y_u = 2/3$  is the up-quark hypercharge, the hypercharge of the neutral seesaw partner of the up quark, so the floor is the lightest-neutrino mass raised to the fourth power, weighted by the hypercharge of the partner it pairs with through the seesaw.

The entropy integral is the logarithm of the ratio of two Gaussian widths, the Planck width  $M_{\text{P}}^2$  and the vacuum floor  $\sqrt{\rho_\Lambda}$ ,

$$\int \gamma_M = \ln \frac{M_{\text{P}}^2 \sqrt{2\pi + r_{23}}}{\sqrt{\rho_\Lambda}} = 139.2537, \quad (\text{C.88})$$

with  $r_{23} = 3/(10\sqrt{3})$  the hypercharge-trace hierarchy ratio. The statistical-mechanical origin is the association entropy of two Gaussians. The Planck anchor is a Gaussian of variance  $M_{\text{P}}^2$  and the vacuum floor a Gaussian of variance  $\sqrt{\rho_\Lambda}$ ; a Gaussian of variance  $\sigma^2$  has differential entropy  $H(\sigma) = \frac{1}{2} \ln(2\pi e \sigma^2)$ , and the association of the two ends, twice their entropy difference plus the entropy-minimum distance  $\ln \sqrt{2\pi + r_{23}}$ , gives

$$\int \gamma_M = 2[H(M_{\text{P}}) - H(\sqrt{\rho_\Lambda})] + \ln \sqrt{2\pi + r_{23}} = \ln \frac{M_{\text{P}}^2 \sqrt{2\pi + r_{23}}}{\sqrt{\rho_\Lambda}}. \quad (\text{C.89})$$

The physical content is the Boltzmann entropy of the scale flow. The scale flow carries two Gaussian measures, one at the Planck anchor of width  $M_{\text{P}}^2$  and one at the vacuum floor of width  $\sqrt{\rho_\Lambda}$ , and the number of available states between the two ends of the emergence window is the ratio of the two widths,  $W = M_{\text{P}}^2 \sqrt{2\pi + r_{23}} / \sqrt{\rho_\Lambda}$ . The factor  $\sqrt{2\pi + r_{23}}$  is the width of the correlation window between the two ends, the Euclidean period  $2\pi$  dressed by the hypercharge-trace hierarchy ratio  $r_{23}$ , so the entropy is the Boltzmann logarithm  $S = \ln W$ , and  $W$  is the window span over which the flow states are counted. The self-similar branch has  $\gamma_M = 0$  (no entropy production, the scale-invariant spectrum), and the infrared end is the maximum-entropy state,  $S = \ln W = 139.25$ , the Jaynes maximum-entropy endpoint of the duality emergence. This single number is the bridge between the



ultraviolet Planck anchor and the infrared vacuum floor. The Hubble rate is the infrared endpoint of the emergence window,

$$H_0 = M_{\text{P}} \sqrt{\pi} e^{-\int \gamma_M}. \quad (\text{C.90})$$

The dark-energy fraction is

$$\Omega_\Lambda = \frac{\rho_\Lambda}{3H_0^2 M_{\text{P}}^2} = \frac{2}{3} + \frac{r_{23}}{3\pi} = 0.68504. \quad (\text{C.91})$$

The algebra is exact. With  $H_0^2 = M_{\text{P}}^2 \pi e^{-2\int \gamma_M}$  and  $e^{-2\int \gamma_M} = \rho_\Lambda / (M_{\text{P}}^4 (2\pi + r_{23}))$ , the ratio is  $\rho_\Lambda / (3H_0^2 M_{\text{P}}^2) = (2\pi + r_{23}) / (3\pi) = 2/3 + r_{23}/(3\pi)$ , the maximum-entropy closure  $2/3$  plus the hypercharge-trace correction  $r_{23}/(3\pi)$ . The microwave temperature is the photon floor set by the lightest neutrino,

$$T_{\text{CMB}} = \frac{m_{\nu_1} r_{12}}{\pi} (1 - \tau \Delta_s), \quad (\text{C.92})$$

with  $r_{12} = 3/10$  and  $\Delta_s = 1/2$  the scalar conformal weight. The physical content is that the lightest fermion sets the photon temperature, the neutrino mass scaled by the hypercharge ratio  $r_{12} = m_1/m_2 = (N_L - N_R) / \sum Y^2$ , the pure content ratio of the chiral difference over the hypercharge capacity, the geometric factor  $1/\pi$ , and the chiral correction  $1 - \tau \Delta_s$  with  $\Delta_s = (d-2)/2 = 1/2$  the scalar conformal weight, the same  $\Delta_s$  that enters the proton-mass correction of Section C.14. The redshift equals the spectrum, in the sense that the photon temperature is the lightest neutrino mass scaled by the content ratio and the period factor.

The primordial spectral tilt is the torsion modulus times the scalar-vector weight ratio,

$$1 - n_s = \tau \cdot \frac{7}{4} = 0.035, \quad (\text{C.93})$$

the  $7/4$  being the scalar-vector mode-weight ratio of the coarse-graining window at the microwave scale. The primordial curvature amplitude is the spin- $\frac{1}{2}$  Gaussian zero-point suppressed by the window period,

$$\Delta_R^2 = \frac{1}{2} \left( \frac{1}{2\pi} \right)^2 e^{-2\pi k L_{\text{CMB}}} (1 - \tau \kappa) = 2.10 \times 10^{-9}, \quad (\text{C.94})$$

with  $\Delta_0^2 = (1/2)(1/2\pi)^2$  the spin- $\frac{1}{2}$  zero-point,  $e^{-2\pi k L_{\text{CMB}}}$  the window suppression, and  $(1 - \tau \kappa)$  the chiral-squash correction, with  $k L_{\text{CMB}} = k L (1 - \tau/4)$  the window at the microwave pivot, so the amplitude closes without inflation.

#### C.14 QCD and the mass gap

The QCD sector closes in three steps. First the condensate energy of the long-root is

$$\Delta E = \frac{1}{8} M_{\text{G}}, \quad (\text{C.95})$$

the curvature gap  $K = 8/3$  times the conformal coupling  $\xi = 1/8$ . The generator mass  $m_{\text{gen}} = g_2(2\tau) M_{\text{G}} / \sqrt{2}$  sets the running initial condition, and the two-loop QCD running with the top matching gives  $\Lambda_{\text{QCD}}$ . The running uses the two-loop gauge beta functions

of Section C.5, whose coefficients are derived there from the group-theoretic content, the Casimir and Dynkin indices of the framework's representations, so no coefficient is imported from a table. The running equation itself,  $\beta_g = g^3/(16\pi^2)(b_i + B_{ij}g_j^2/(16\pi^2) - A_i y_t^2/(16\pi^2))$ , is the standard form of the renormalisation group, and it enters at this single point as the statement that the gauge coupling evolves logarithmically with the scale. The framework's independent content is the initial condition  $g_3(M_G)$ , the geometric Yukawa  $y_0 = 1$ , and the coefficients  $b_i$  and  $B_{ij}$  of Section C.5. The two-loop terms shift  $\Lambda_{\text{QCD}}$  at the level of a few percent, and the one-loop content of the running is fixed entirely by the framework's coefficients. Second the topological gap is the  $l = 2$  scalar mode of  $RP^3$ , whose eigenvalue is  $\lambda_{\text{glue}} = 8/L^2 > 0$ , the lowest colour-singlet level. Third the glueball tower is the two-gluon bound-state spectrum,

$$\lambda = 2\lambda_{\text{gluon}} + C_2(J) + n(N_g\xi), \quad N_g\xi = 1, \quad (\text{C.96})$$

with  $\lambda_{\text{gluon}} = 4/L^2$  the Killing eigenvalue. The lightest glueball is

$$m_G = 8\Lambda_{\text{QCD}}, \quad (\text{C.97})$$

and the tower ratios are  $2^{++}/0^{++} = \sqrt{2}$ ,  $0^{++*}/0^{++} = 3/2$ ,  $0^{-+}/0^{++} = \sqrt{17/8}$ . The proton mass is the constituent-quark content with the conformal-duality correction,

$$m_p = N_c \Delta_f \left(1 - \frac{1}{N_g^2 \Delta_s}\right) \Lambda_{\text{QCD}} = \frac{279}{64} \Lambda_{\text{QCD}}, \quad (\text{C.98})$$

where  $1/(N_g^2 \Delta_s) = 1/32 = \xi/4$  follows from the two identities  $N_g\xi = 1$  and  $N_g\Delta_s = 2(d-1)$ . The coefficient is the product of the three constituent quarks,  $N_c = 3$ , the fermion conformal weight  $\Delta_f = d/2 = 3/2$ , and the conformal-duality factor  $31/32$ , so  $(9/2)(31/32) = 279/64$ . The scheme bracket  $(1 + \tau\kappa \sum Y^2 \Delta_s) = (1 + 5\tau\kappa/3)$  is the correction between the constituent-quark scale and the  $\overline{\text{MS}}$  scale, with  $\sum Y^2 \Delta_s = (10/3)(1/2) = 5/3$  the hypercharge capacity times the scalar conformal weight. The string tension and the deconfinement temperature are

$$\sigma = \frac{\lambda_{\text{TT}}}{\pi} \Lambda_{\text{QCD}}^2 = \frac{14}{\pi} \Lambda_{\text{QCD}}^2, \quad T_d = \frac{\lambda_{\text{vector}}}{N_c} \Lambda_{\text{QCD}} = \frac{4}{3} \Lambda_{\text{QCD}}, \quad (\text{C.99})$$

with the self-consistent ratio  $\sigma/T_d^2 = (14/\pi)(9/16) = 5/2$ .

### C.15 Primordial nucleosynthesis

The six constants of the weak-rate freeze-out are derived from the framework content, not taken from observation. The Fermi constant is  $G_F = 1/(\sqrt{2}v^2)$  from the closed electroweak scale. The axial coupling is the conformal-weight form

$$g_A = \frac{N_g \Delta_s}{\pi} = \frac{2(d-1)}{\pi} = \frac{4}{\pi}. \quad (\text{C.100})$$

This is the framework's axial coupling, the generator count  $N_g = 2(d-1) = 8$  times the scalar conformal weight  $\Delta_s = (d-2)/2 = 1/2$  over the Euclidean period  $\pi$ , the

same conformal-gauge content that closes the gauge sector of Section C.4. It is not the nucleon axial charge of the standard model: it is the conformal-weight axial coupling of the framework’s content, used in the framework’s own weak-rate freeze-out, not substituted into an external nucleon matrix element. The electromagnetic self-energy is  $\Delta_{\text{EM}} = (1 - 1/(2\pi))\alpha\Lambda_{\text{QCD}}$ , and the radiative correction is  $\delta_R = 1 + (1 - \tau)/(8\pi)$ . The neutron-proton mass difference is the quark mass difference minus the self-energy,

$$\Delta m_{np} = (m_d - m_u) - \Delta_{\text{EM}}, \quad (\text{C.101})$$

with the quark masses from the ladder of Section C.7. The neutron lifetime is

$$\tau_n = \frac{2\pi^3}{G_F^2 |V_{ud}|^2 (1 + 3g_A^2) m_e^5 f \delta_R}, \quad (\text{C.102})$$

with  $|V_{ud}| = \sqrt{1 - |V_{us}|^2}$  and  $f$  the phase-space integral. The phase-space factor  $f = \int F(Z, W) W p(W_0 - W)^2 dW$  is the framework’s own spectral integral of the weak-decay kinematics, the Fermi Coulomb function  $F(Z, W)$  (with  $Z = 1$  for the proton) times the phase-space weight  $W p(W_0 - W)^2$  integrated over the electron energy, computed numerically to full precision, with  $W_0 = \Delta m/m_e$  the endpoint; it is a pure-kinematics spectral sum with no free parameter and no imported decay constant. The freeze-out temperature solves  $\Gamma_{\text{weak}}(T) = H(T)$ , and the helium yield is

$$Y_p = \frac{2(n/p)_{\text{BBN}}}{1 + (n/p)_{\text{BBN}}}, \quad (n/p)_{\text{BBN}} = e^{-\Delta m/T_f} e^{-t/\tau_n}. \quad (\text{C.103})$$

The effective neutrino species is  $N_{\text{eff}} = 3 + \sqrt{3}/(2\pi)^2$ . All six constants close through the conformal-weight form, the  $2\pi$  Euclidean period, and the torsion content ratio, with no QCD loop and no relativistic correction.

## D Theoretical sensitivity

### D.1 Sensitivity of the hierarchy

The central values of this paper are exact in the sense of Section 2.3: no free parameter and no fitted coefficient enter the chain. Exactness of the central values, however, does not by itself quantify how the outputs respond to the inputs and conventions that carry the chain; this appendix records that response in full. The inputs separate into four layers. The single observed anchor  $G_N$  carries the experimental relative uncertainty of order  $2 \times 10^{-5}$ , which propagates only into the dimensionful quantities (each with elasticity +1) and leaves every dimensionless quantity unchanged — the  $M_P$ -rescaling invariance. The structural numbers 2,  $\pi$ ,  $3/2$ ,  $1/8$  and the geometric factors are exact. The remaining two layers — the window-capacity fixed point  $kL$  and the torsion modulus  $\tau$  — are the derived intermediates whose defining conventions carry the genuine theoretical sensitivity, and it is these that the analysis quantifies.

Table 3 gives the elasticity matrix  $d \ln O / d \ln p$  for the principal outputs and the three inputs  $p = kL, \tau, M_P$ , evaluated by central finite difference on the closed forms of Appendix C. The entries for  $p = kL$  are the load-bearing result. The electroweak scale  $v$ ,

| Quantity $O$     | Elasticity $d \ln O / d \ln p$ |            |           |
|------------------|--------------------------------|------------|-----------|
|                  | $p = kL$                       | $p = \tau$ | $p = M_P$ |
| $M_G$            | -1.0                           | 0.0        | +1.0      |
| $g_2(M_G)$       | -2.5                           | 0.0        | 0         |
| $m_t/m_c$        | +5.0                           | -0.1       | 0         |
| $v$              | -32.3                          | -0.1       | +1.0      |
| $m_e$            | -49.9                          | -0.1       | +1.0      |
| $m_{\nu_3}$      | -64.7                          | -1.1       | +1.0      |
| $\rho_\Lambda$   | -261.6                         | -4.5       | +4.0      |
| $H_0$            | -129.7                         | -2.2       | +1.0      |
| $\Omega_\Lambda$ | 0.0                            | 0.0        | 0         |

**Table 3.** The elasticity matrix: the percentage change of each output per one-percent change of  $kL$ ,  $\tau$ , or  $M_P$ . The large negative entries in the  $kL$  column reflect the exponential hierarchy  $e^{-4\pi kL}$ ,  $e^{-20kL}$ , and their powers.

| Convention (1% shift)                 | $\rightarrow kL$ | $\rightarrow v$ | $\rightarrow m_e$ | $\rightarrow \rho_\Lambda$ |
|---------------------------------------|------------------|-----------------|-------------------|----------------------------|
| Kernel exponent $(1 - y)^p$ , $p = 2$ | +0.8%            | -26%            | -40%              | -212%                      |
| Threshold $4/27$                      | -1.0%            | +32%            | +50%              | +262%                      |
| Window $kL$                           | —                | -32%            | -50%              | -262%                      |

**Table 4.** The convention chain: how a one-percent shift of each upstream convention propagates into the fixed point  $kL$  and then into the hierarchy.

the electron mass  $m_e$ , the heaviest neutrino mass  $m_{\nu_3}$ , the dark-energy density  $\rho_\Lambda$  and the Hubble rate  $H_0$  all carry large negative elasticities, because they are built from exponentials of  $kL$ :  $v \propto e^{-4\pi kL}$ ,  $m_e \propto e^{-20kL}$ ,  $m_{\nu_3} \propto v^2$ ,  $\rho_\Lambda \propto m_{\nu_1}^4$ ,  $H_0 \propto m_{\nu_1}^2$ . A one-percent shift of  $kL$  therefore moves  $v$  by -32%,  $m_e$  by -50%, and  $\rho_\Lambda$  by -262%. The mass ratios carry only order-unity elasticities, and the dark-energy fraction  $\Omega_\Lambda = 2/3 + r_{23}/(3\pi)$  is a pure content ratio whose elasticity is exactly zero — the hierarchy lives in the exponentials of  $kL$ , not in  $\Omega_\Lambda$ .

The elasticity into  $kL$  traces back to the three conventions that fix the fixed point, through the spectral-pole condition  $V_3 \Pi^2/(32\pi^2) = 4/27$  of Section C.1. Table 4 records this chain. The kernel exponent is the transverse-traceless kernel  $K_{\text{TT}} = (1 - y)^2$  regarded as  $p = 2$  of the one-parameter family  $(1 - y)^p$ ; the threshold  $4/27$  is the maximum of the mass-weighted kernel  $y(1 - y)^2$ . A one-percent shift of the kernel exponent moves the threshold by  $p \ln(p/(p + 1)) = -0.81\%$ , the fixed point by +0.81% (the elasticity  $d \ln kL / d \ln(4/27)$  is exactly -1), and  $v$  by -26%. The three conventions therefore sit at the top of a steep amplification chain.

The status of the three conventions is not uniform, and it is stated here rather than left implicit. The kernel  $K_{\text{TT}} = (1 - y)^2$  is the standard spectral density of the transverse-traceless projection of the improved energy-momentum tensor:  $K_{\text{TT}} = k^4/(k^2 + m^2)^2$ ,

and the substitution  $y = m^2/(k^2 + m^2)$  turns it into  $(1 - y)^2$ . This is an established consequence of the spin-2 Ward identity — the kernel vanishes at  $m^2 = 0$ , as a massless graviton is transverse-traceless — and is borrowed from general relativity, not introduced by the framework. The threshold  $4/27$  is the exact maximum of  $y(1 - y)^2$  at  $y = 1/3$ , a calculus identity; the two choices that precede it — the mass weight  $y$ , which measures how far the torsion has activated the TT channel, and the selection of the maximum as the normalisation — are physically motivated conventions, not unique deductions. The window capacity  $2\pi kL^4$  is a notation rather than an independent spectral sum: it enters the bare field equation as  $1/(2\pi kL^4)$  and the hypercharge screening as  $2\pi kL^4$ , so it cancels exactly and drops out of  $\tau$ . The torsion modulus is therefore fixed by the content ratio alone,  $\tau = (N_L - N_R)/(N_f \sum Y^2) = 1/50$ , an integer identity, and does not depend on the specific value of  $2\pi kL^4$ . This is the precise sense in which the hierarchy is exact but convention-sensitive: the central values carry no free parameter, yet they rest on three upstream conventions, of which one — the transverse-traceless kernel — is inherited from general relativity, one — the threshold  $4/27$  — is an exact extremum with a conventional selection, and one — the window capacity  $2\pi kL^4$  — is a notation that cancels out of the physical content.

The selection of the maximum can be sharpened, and its status stated precisely. The mass-weighted density  $y(1 - y)^2$  has a unique interior stationary point:  $[y(1 - y)^2]' = (1 - y)(1 - 3y)$  vanishes on  $(0, 1)$  only at  $y = 1/3$ , where the second derivative is  $-2$ , so  $4/27$  is the unique interior extremum, the endpoints  $y = 0, 1$  giving zero. There is no competing interior value: the maximum is the only non-trivial scale-invariant datum carried by the kernel. It is also the point of maximal activation of the transverse-traceless channel — the mass fraction  $y = 1/3$  balancing the two kinetic fractions  $(2/3)^2$  — which is the point at which the massless pole appears. This selection is a consequence of the disorder axiom rather than an independent hypothesis: the unbiasedness clause of the axiom, together with the maximum-entropy theorem of the companion paper [1], fixes the flow to the configuration of maximal spectral weight, and the maximum of  $y(1 - y)^2$  is exactly that configuration. The extremum is thus singled out both by uniqueness and by the maximum-entropy theorem of the companion paper, which lifts the selection from a bare convention to a consequence of the single axiom. A third, independent reading places the same number in the content symmetries. The stationary condition  $y = 1/3$  balances one mass fraction against two kinetic fractions,  $(1/3)(2/3)^2$ , which is exactly the content of a spin-2 mode: one mass direction (the longitudinal activation by the torsion mass) against the two transverse polarisations carried by the two powers of  $k^2$  in the kernel. The value  $4/27$  is thus the content datum of the spin-2 channel, of the same kind as the integer content identities  $N_L = N_g = 8$  and  $\sum Y^2 = 10/3$  that fix the other structural numbers. Uniqueness, maximum entropy, and content symmetry therefore coincide on the same extremum, which is why the threshold is not a free convention.

The kernel is likewise unique rather than chosen. The transverse-traceless projection of a spin-2 mode is forced by the Ward identity (transversality) together with tracelessness, and its spectral density is  $K_{\text{TT}} = k^4/(k^2 + m^2)^2$ ; the two powers of  $k^2$  are the two transverse polarisations and the denominator is the propagator squared, so with  $y = m^2/(k^2 + m^2)$

| Convention blur        | $\rightarrow kL$ | $\rightarrow v$ | $\rightarrow m_e$ | $\rightarrow \rho_\Lambda$ |
|------------------------|------------------|-----------------|-------------------|----------------------------|
| $y \rightarrow 1.10 y$ | +0.73%           | -23%            | -36%              | -190%                      |
| $y \rightarrow 1.30 y$ | +6.1%            | -196%           | -303%             | -1590%                     |
| $p \rightarrow 1.01 p$ | +0.81%           | -26%            | -41%              | -212%                      |

**Table 5.** The actual error band of the hierarchy under a stated prior blur of the mass fraction  $y$  or of the kernel exponent  $p$ . The flatness of the extremum softens only the first link ( $y \rightarrow$  threshold); the threshold-to- $kL$  and  $kL$ -to-output links remain first-order, so a 10% blur of  $y$  still moves  $v$  by about 23%.

the kernel is exactly  $(1 - y)^2$ . No other function is transverse, traceless, and of spin 2, so  $(1 - y)^2$  is unique given the spin-2 requirement, and is inherited from general relativity rather than selected.

The flatness of the extremum softens only the first link of the chain, and the remainder stays first-order. Since  $4/27$  is a stationary point, a shift of the mass fraction by  $\Delta y$  moves the threshold only at second order:  $y(1 - y)^2 = 4/27 - (y - 1/3)^2 + \mathcal{O}((y - 1/3)^3)$  locally, the curvature being  $|[y(1 - y)^2]''_{1/3}| = 2$ . But the threshold acts on the fixed point  $kL$  at first order (the elasticity  $d \ln kL / d \ln(4/27) = -1$ ), and  $kL$  acts on the hierarchy at first order through the exponentials of Table 3. The two links downstream of the threshold are therefore not softened by the flatness of the extremum. Table 5 records the actual response of the principal outputs to a stated prior blur of each convention.

The conclusion is therefore the opposite of a free-parameter statement. The threshold is flat, so the convention that fixes it is not sharply localised in  $y$ ; but the hierarchy downstream is exponential, so any residual blur of the convention is amplified by one to two orders of magnitude into the dimensionful output. The framework is exact in its central values and convention-sensitive in its hierarchy, and the error band of Table 5 quantifies that sensitivity under an explicit prior rather than leaving it as a bare elasticity.

The subjective priors of Table 5 can be replaced by an objective constraint on the regularisation scheme. The framework carries two schemes, the smooth Gaussian window — the coarse-graining envelope, the unbiasedness clause of the disorder axiom — and the sharp Litim step. Re-solving the fixed point with the Litim step gives  $kL = 1.09$  against the Gaussian  $kL = 2.49$ , a spread of -56%. The Litim value, however, fails the generation count of Section C.1:  $(1.09)^2 = 1.19 < 1.5$ , so no spinor mode fits the window and the generation count is zero, contradicting the observed three generations. The generation count therefore fixes the scheme: only the Gaussian window gives a window capacity  $(2.49)^2 = 6.22$  large enough to hold the three modes  $n = 0, 2, 4$ . The regularisation scheme is thus not a free convention — it is fixed by the unbiasedness clause of the disorder axiom together with the observed three generations — and its objective blur is zero.

## D.2 The spectral-scale flow and the renormalisation group

The spectral-scale flow of the framework is the evolution of the spectral datum with the coarse-graining scale, parametrised by the trajectory  $L(k) = C/k$  of Section C.1. Its relation to the standard renormalisation group is as follows. Below the emergence scale  $M_G$  the

spectral sums are dominated by the zero modes and the lowest Kaluza–Klein excitations, the regime in which the spectral-sum definition and the standard measure-theoretic formulation coincide, as proved in Appendix C of the companion paper [1]. In this regime the scale flow of the effective couplings reduces to the standard two-loop renormalisation group equations of Section C.5, whose coefficients are the group-theoretic content of the framework rather than imported values. The spectral-scale flow therefore matches the renormalisation group equations in the precise sense that both describe the same low-energy running of the effective couplings in the shared regime: the flow supplies the spectral datum and the boundary condition at  $M_G$ , while the standard equations reproduce the running below  $M_G$  with coefficients fixed by the same content. This is an identification in the shared low-energy regime, not a claim of global equivalence between the two formulations.

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